

ML Hybrid Models Comparison Report

Generated: 2025-12-05T17:59:14.118340

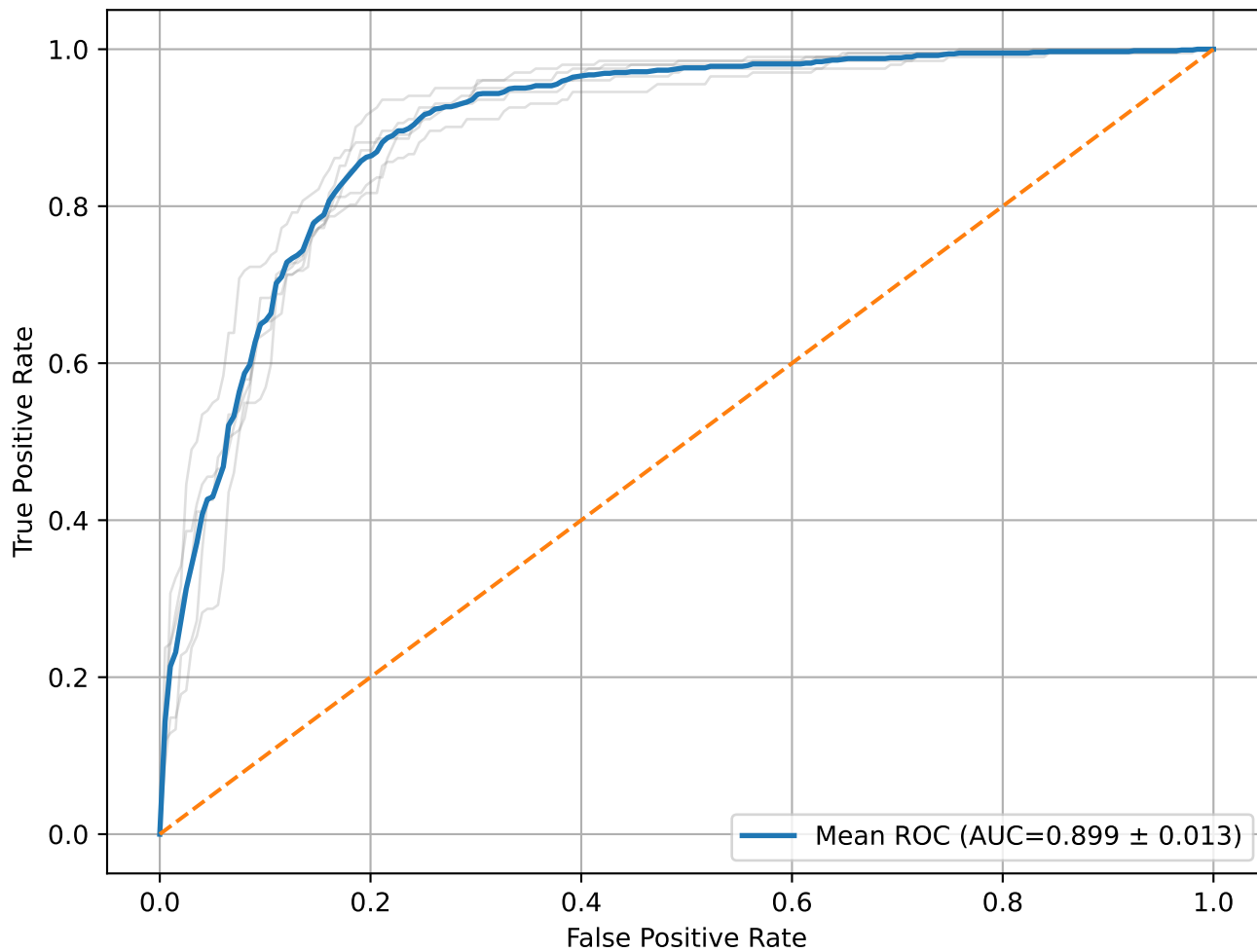
Dataset samples: 2007

Models: logistic, rf, svm, xgb, lgb, dnn, stacking, voting, bagging, pca_svm, ga_xgb, pso_svm, bayes_opt_svms

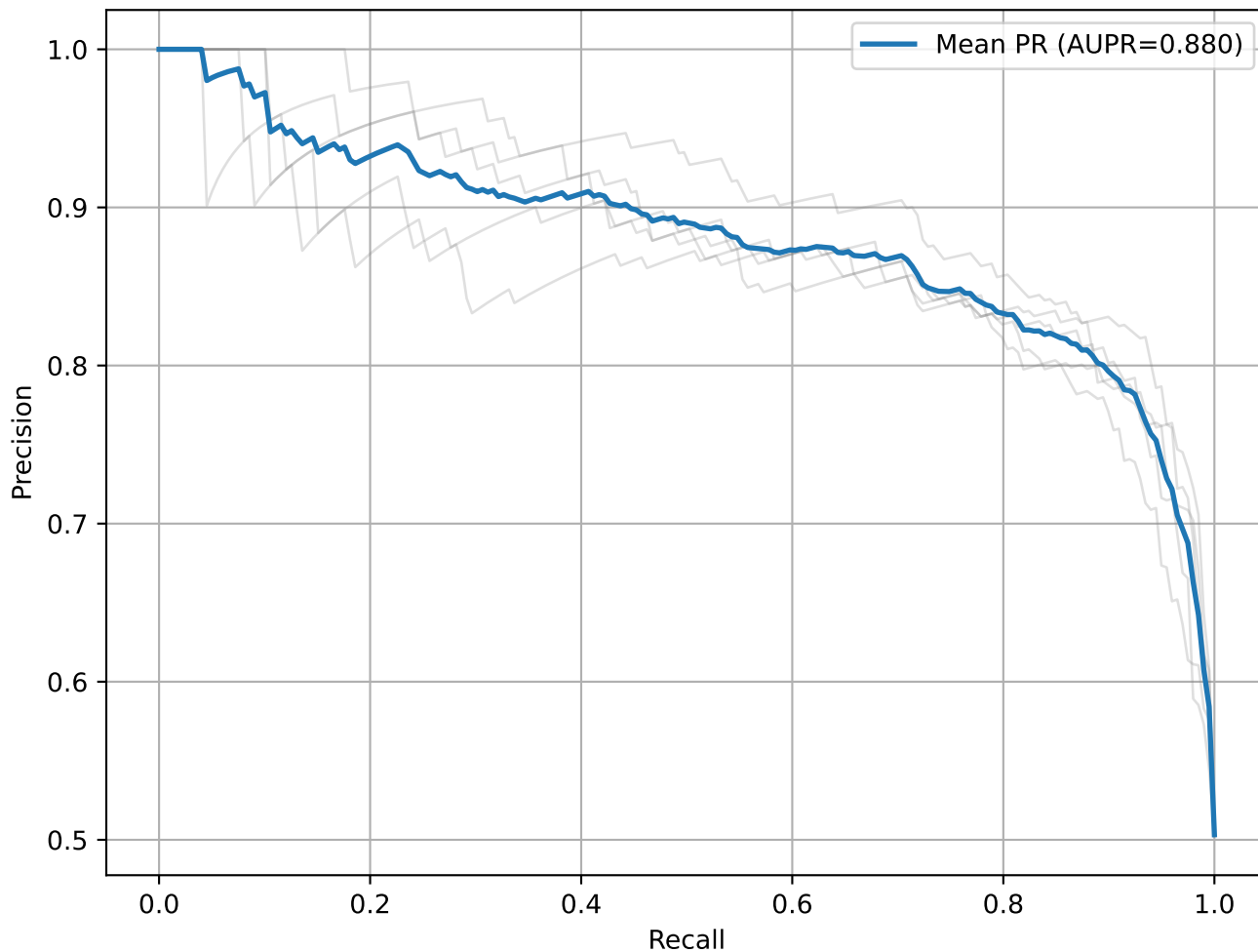
StratifiedKFold: 5 splits

Random seed: 475

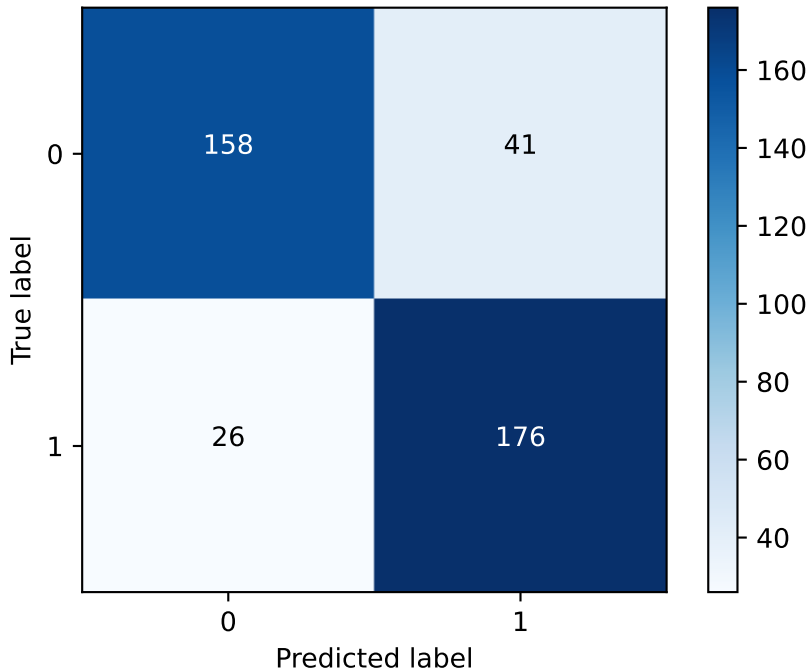
ROC - LOGISTIC



Precision-Recall - LOGISTIC



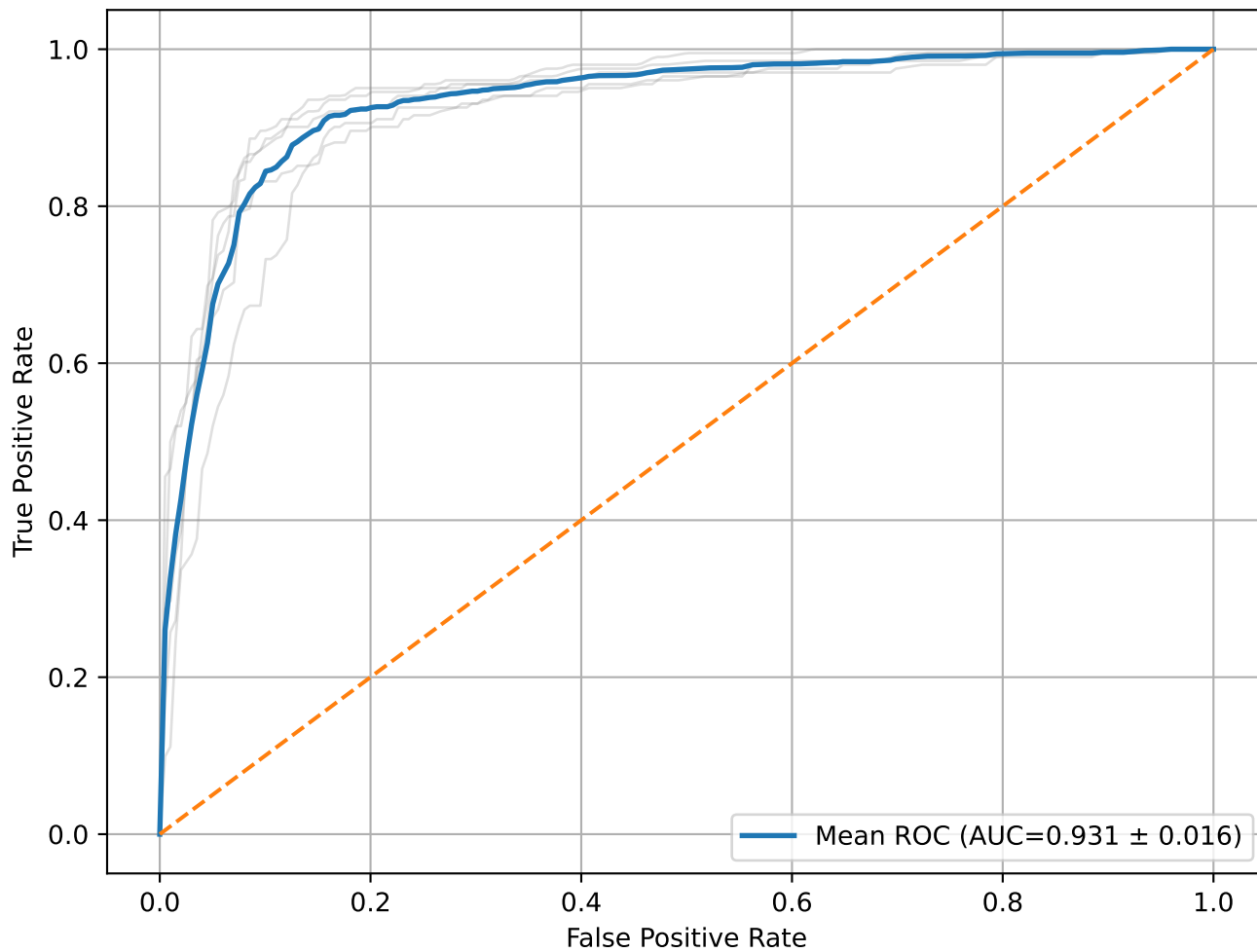
Confusion Matrix - LOGISTIC (last fold)



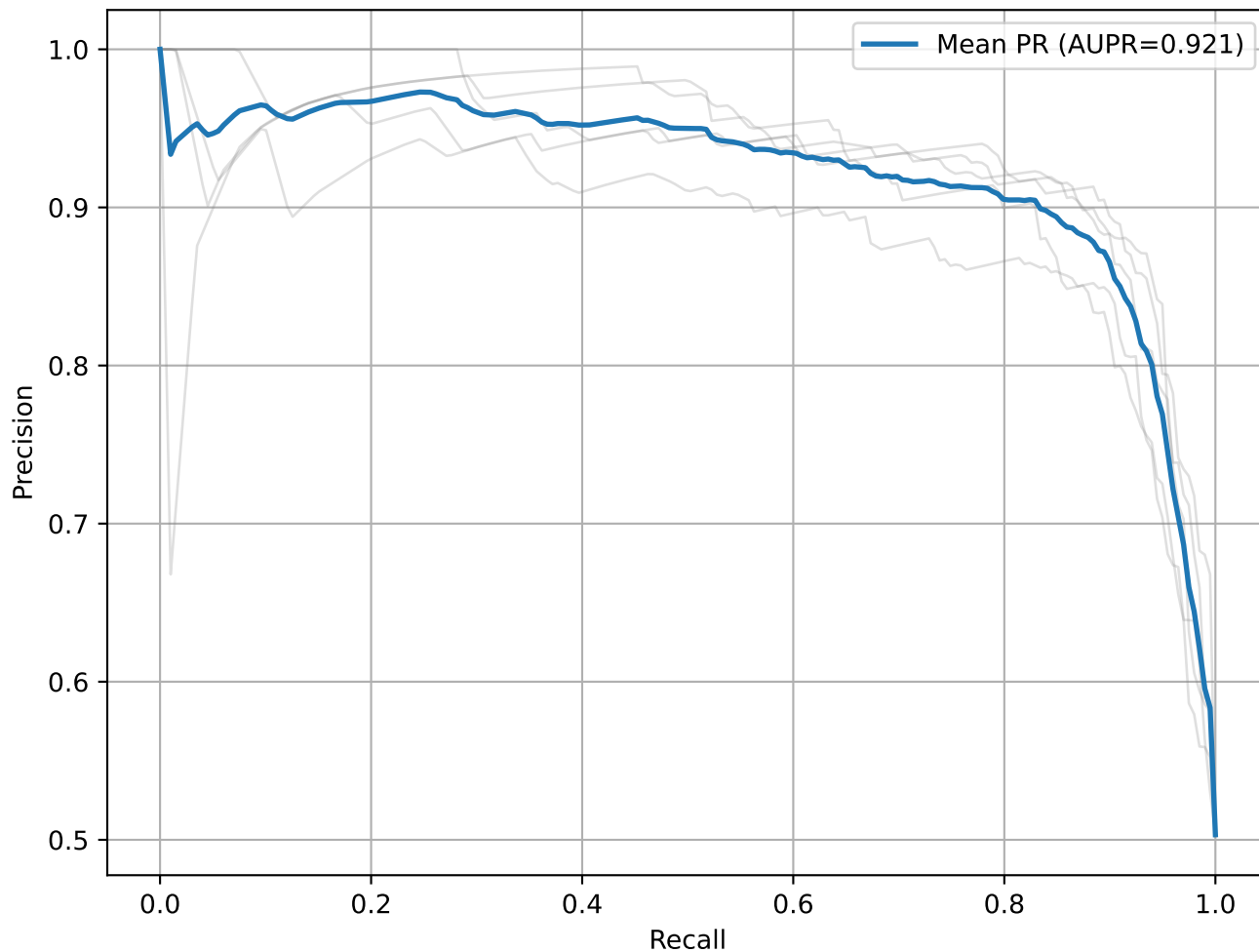
Classification Report - LOGISTIC (last fold)

	precision	recall	f1-score	support
0	0.86	0.79	0.83	199
1	0.81	0.87	0.84	202
accuracy			0.83	401
macro avg	0.83	0.83	0.83	401
weighted avg	0.83	0.83	0.83	401

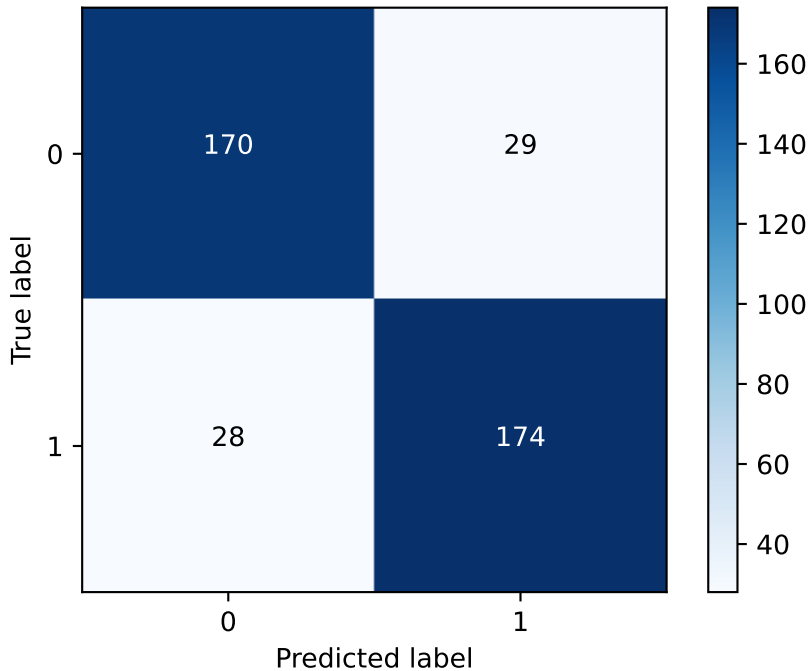
ROC - RF



Precision-Recall - RF



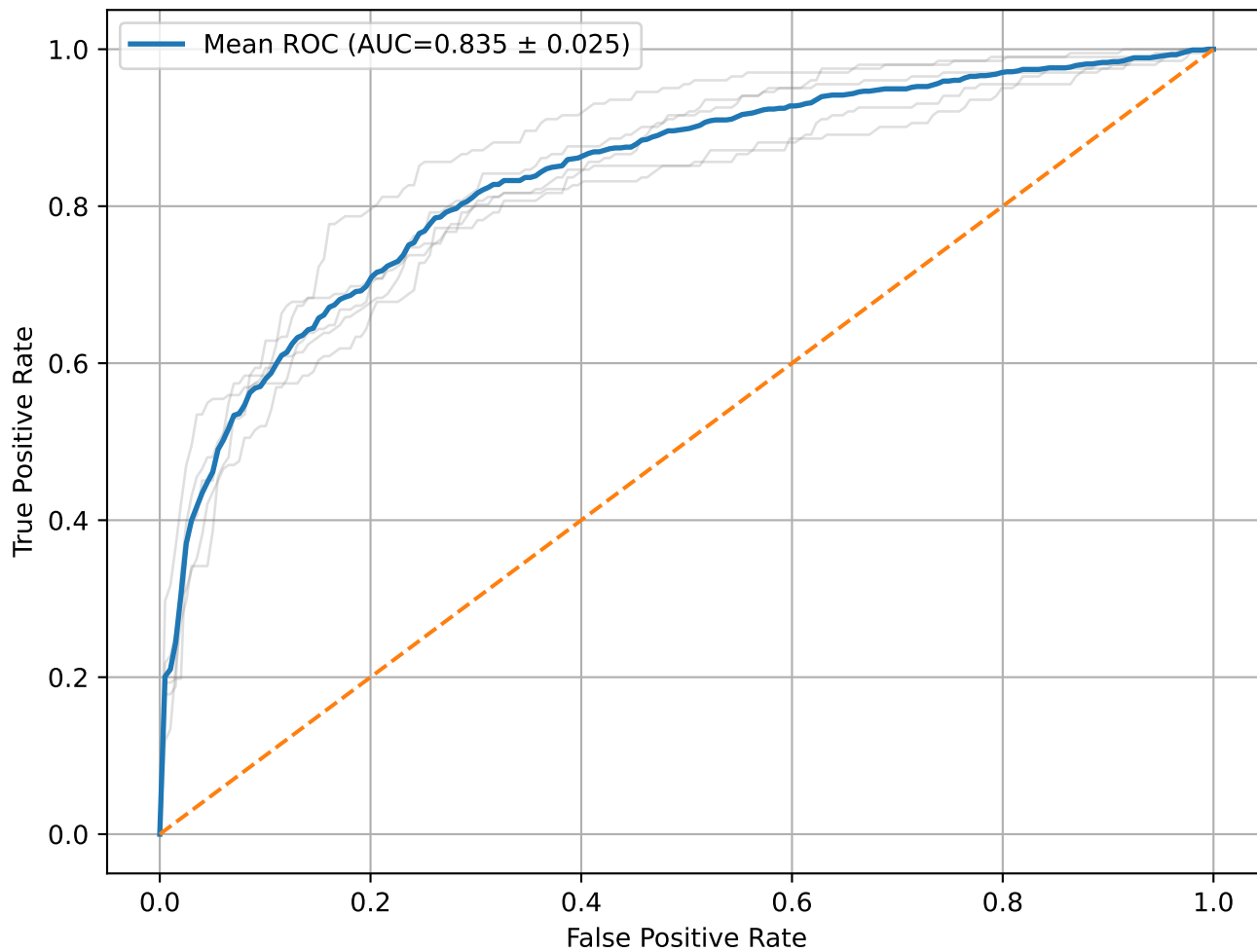
Confusion Matrix - RF (last fold)



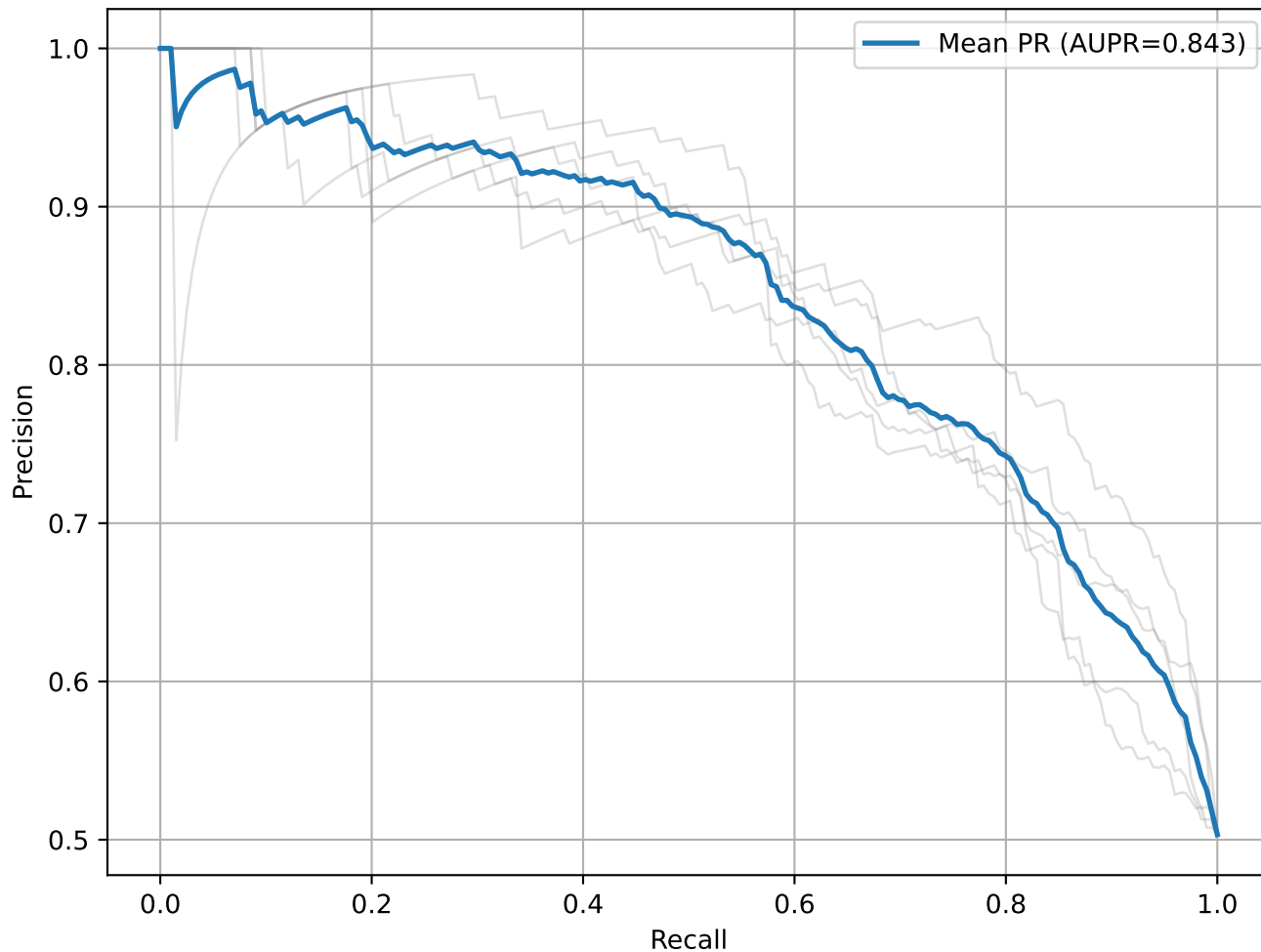
Classification Report - RF (last fold)

	precision	recall	f1-score	support
0	0.86	0.85	0.86	199
1	0.86	0.86	0.86	202
accuracy			0.86	401
macro avg	0.86	0.86	0.86	401
weighted avg	0.86	0.86	0.86	401

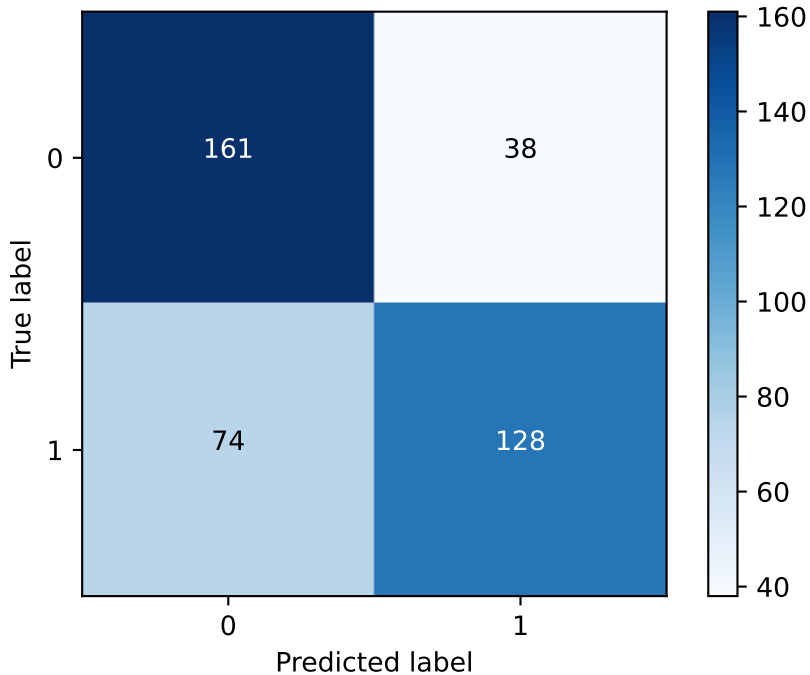
ROC - SVM



Precision-Recall - SVM



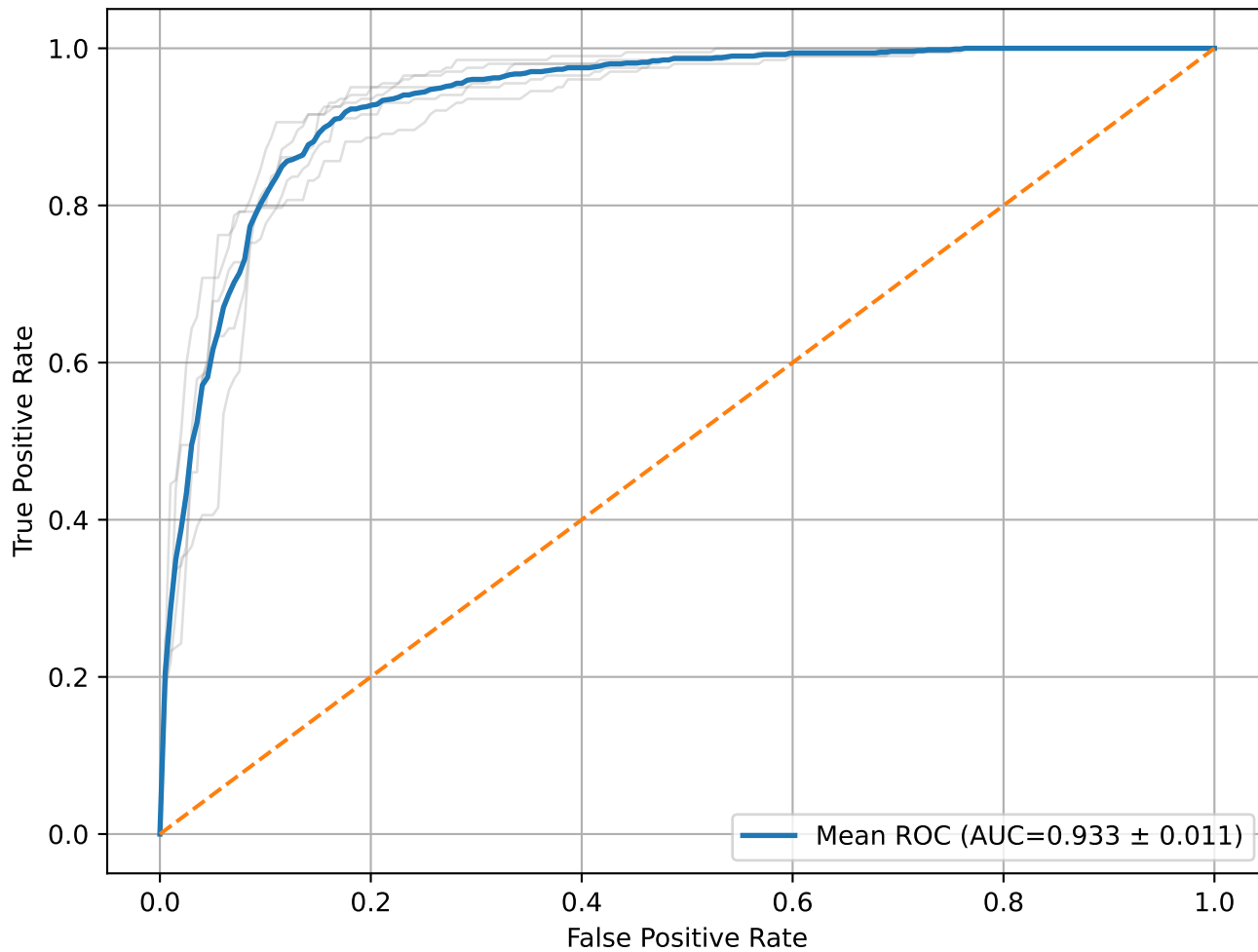
Confusion Matrix - SVM (last fold)



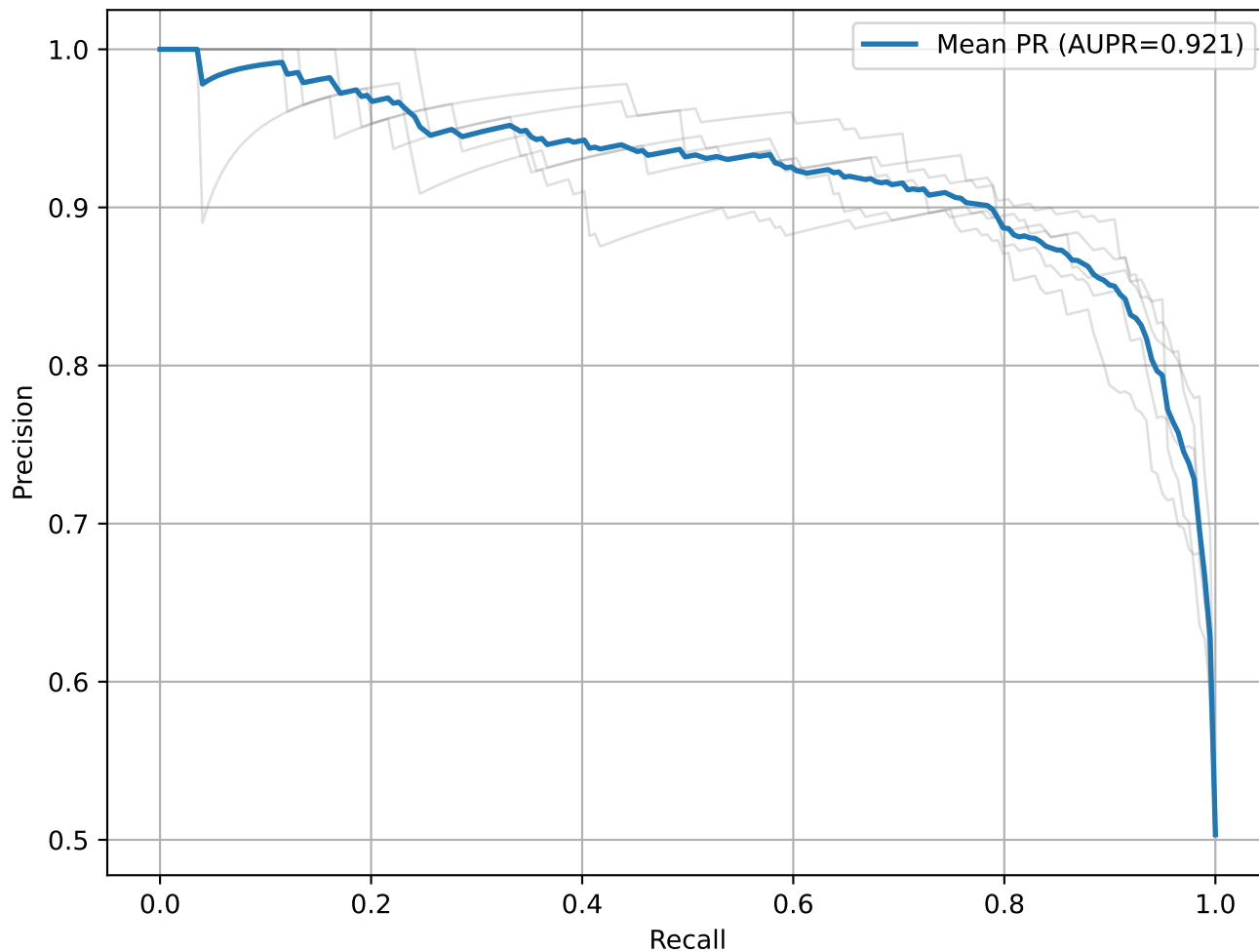
Classification Report - SVM (last fold)

	precision	recall	f1-score	support
0	0.69	0.81	0.74	199
1	0.77	0.63	0.70	202
accuracy			0.72	401
macro avg	0.73	0.72	0.72	401
weighted avg	0.73	0.72	0.72	401

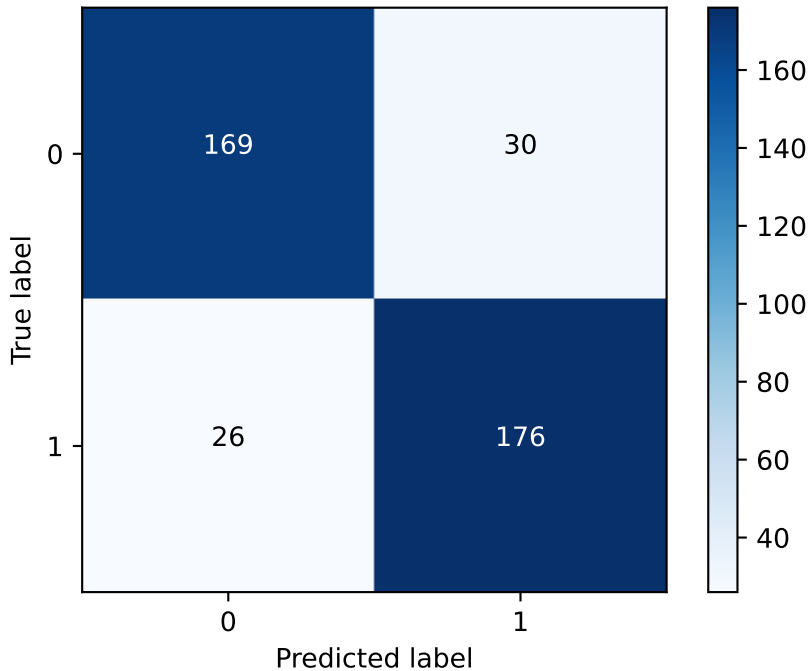
ROC - XGB



Precision-Recall - XGB



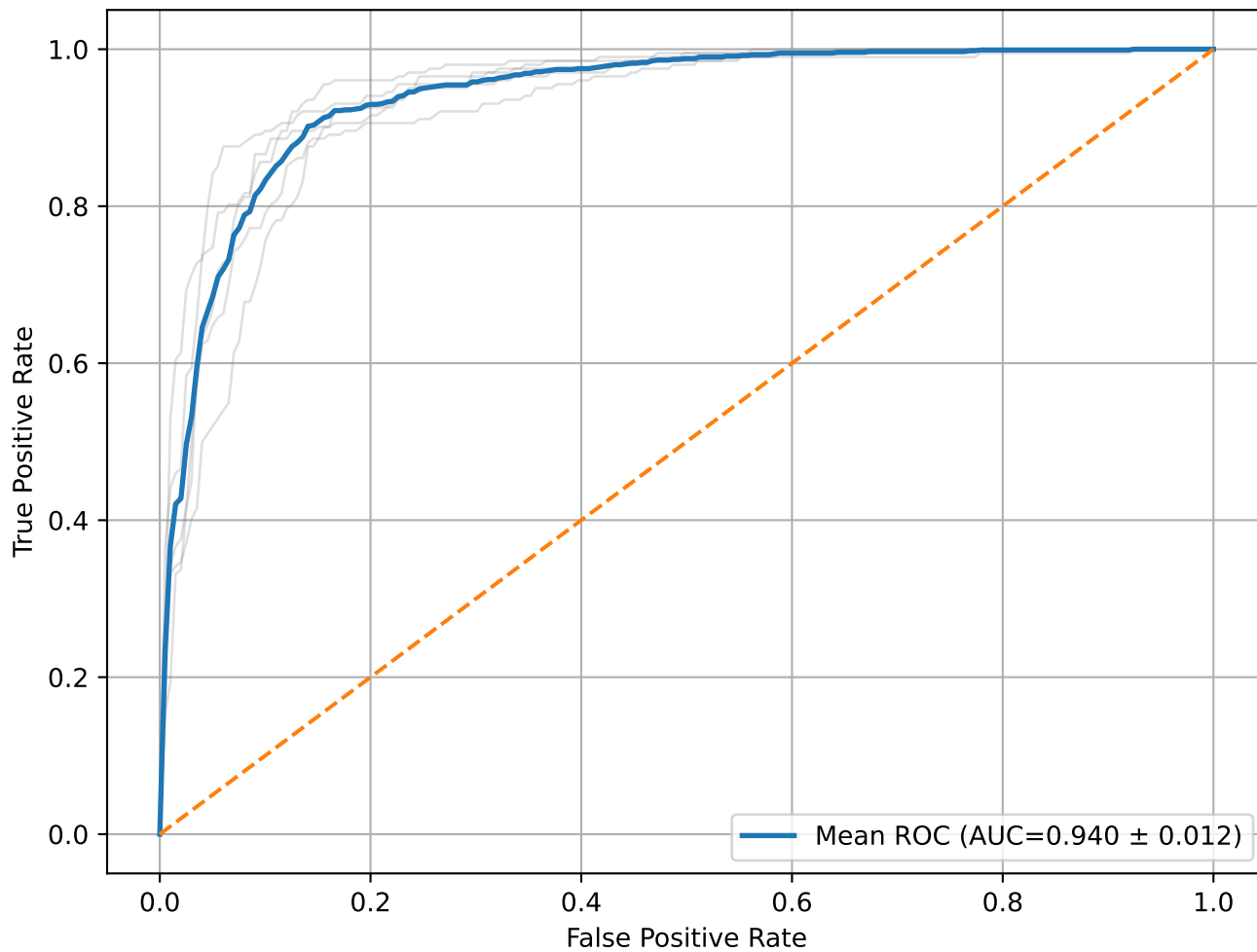
Confusion Matrix - XGB (last fold)



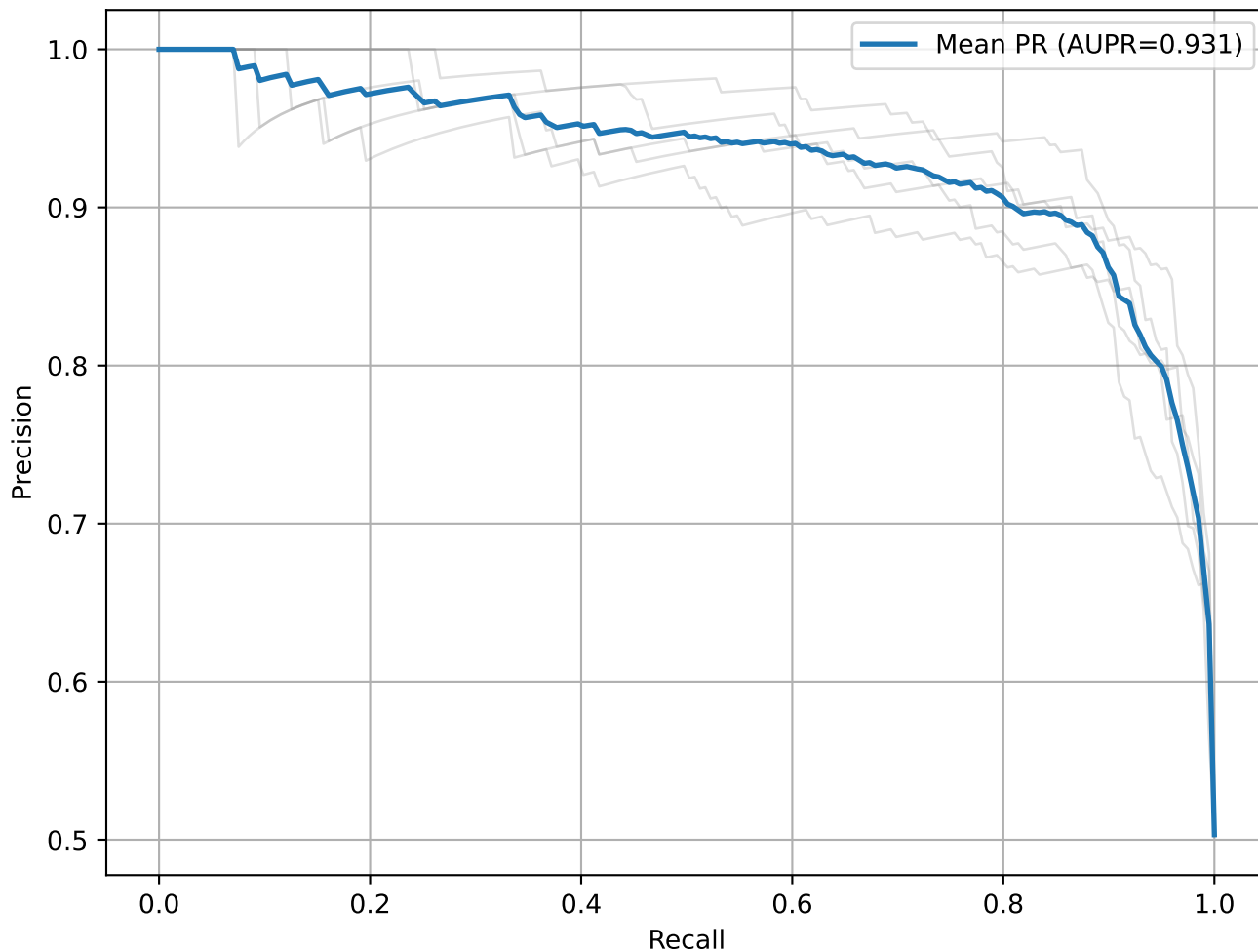
Classification Report - XGB (last fold)

	precision	recall	f1-score	support
0	0.87	0.85	0.86	199
1	0.85	0.87	0.86	202
accuracy			0.86	401
macro avg	0.86	0.86	0.86	401
weighted avg	0.86	0.86	0.86	401

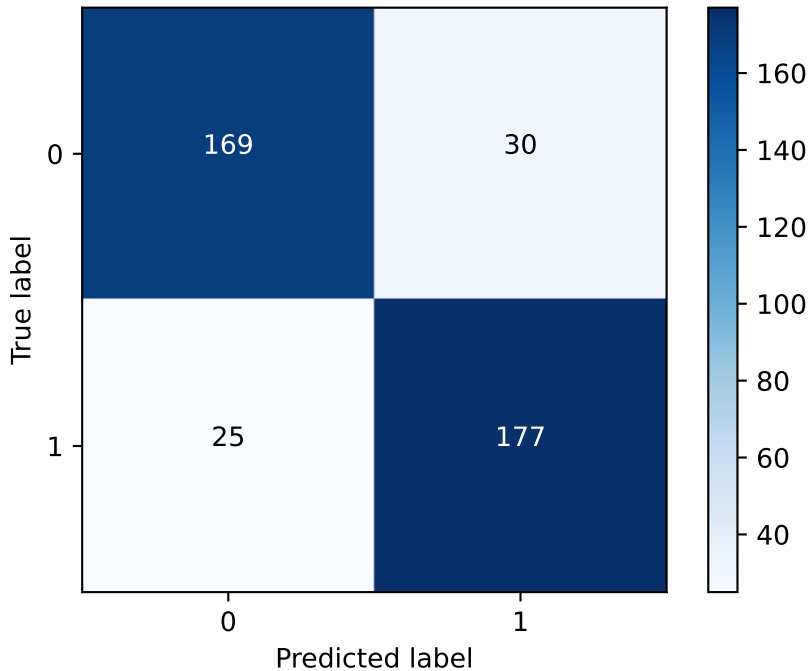
ROC - LGB



Precision-Recall - LGB



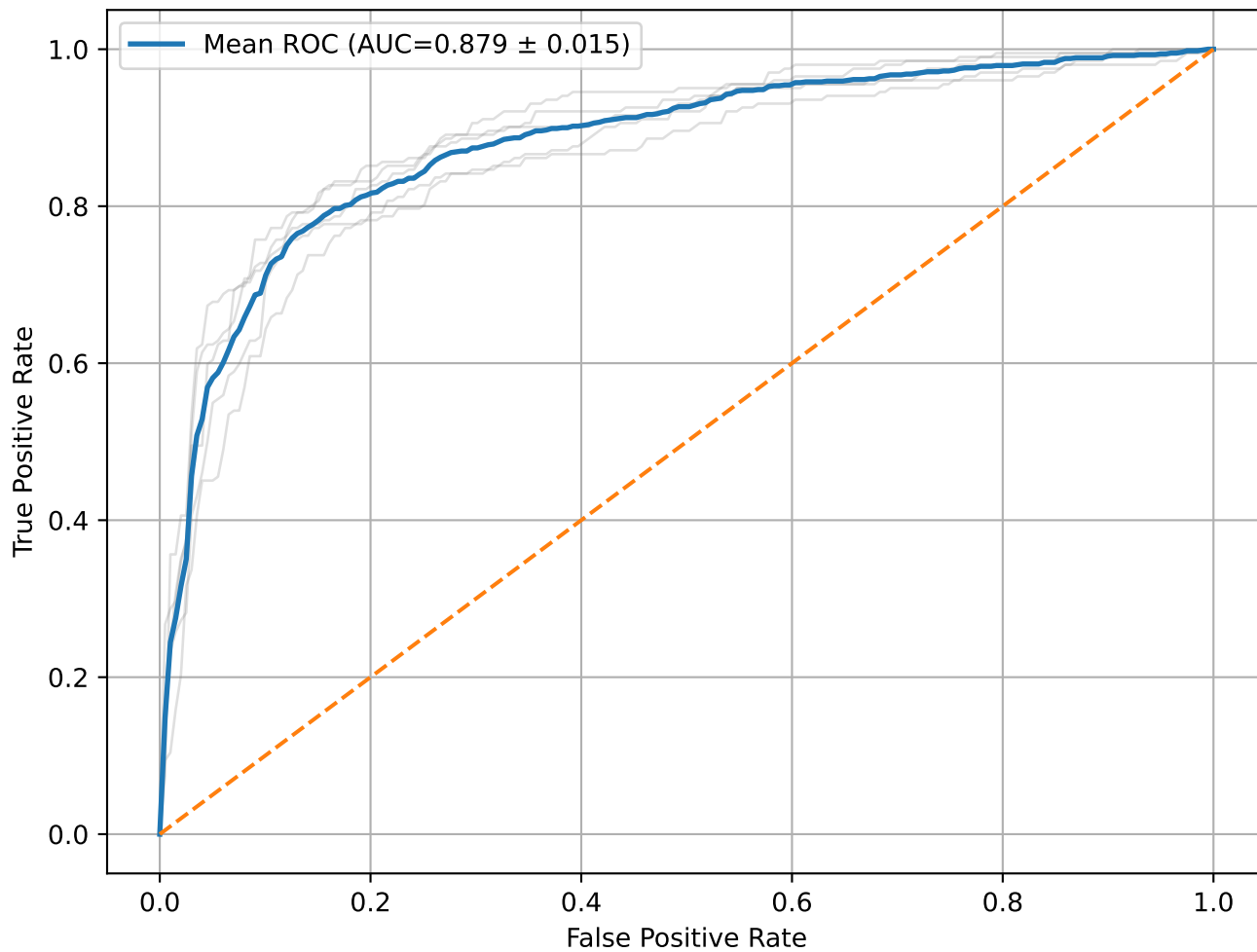
Confusion Matrix - LGB (last fold)



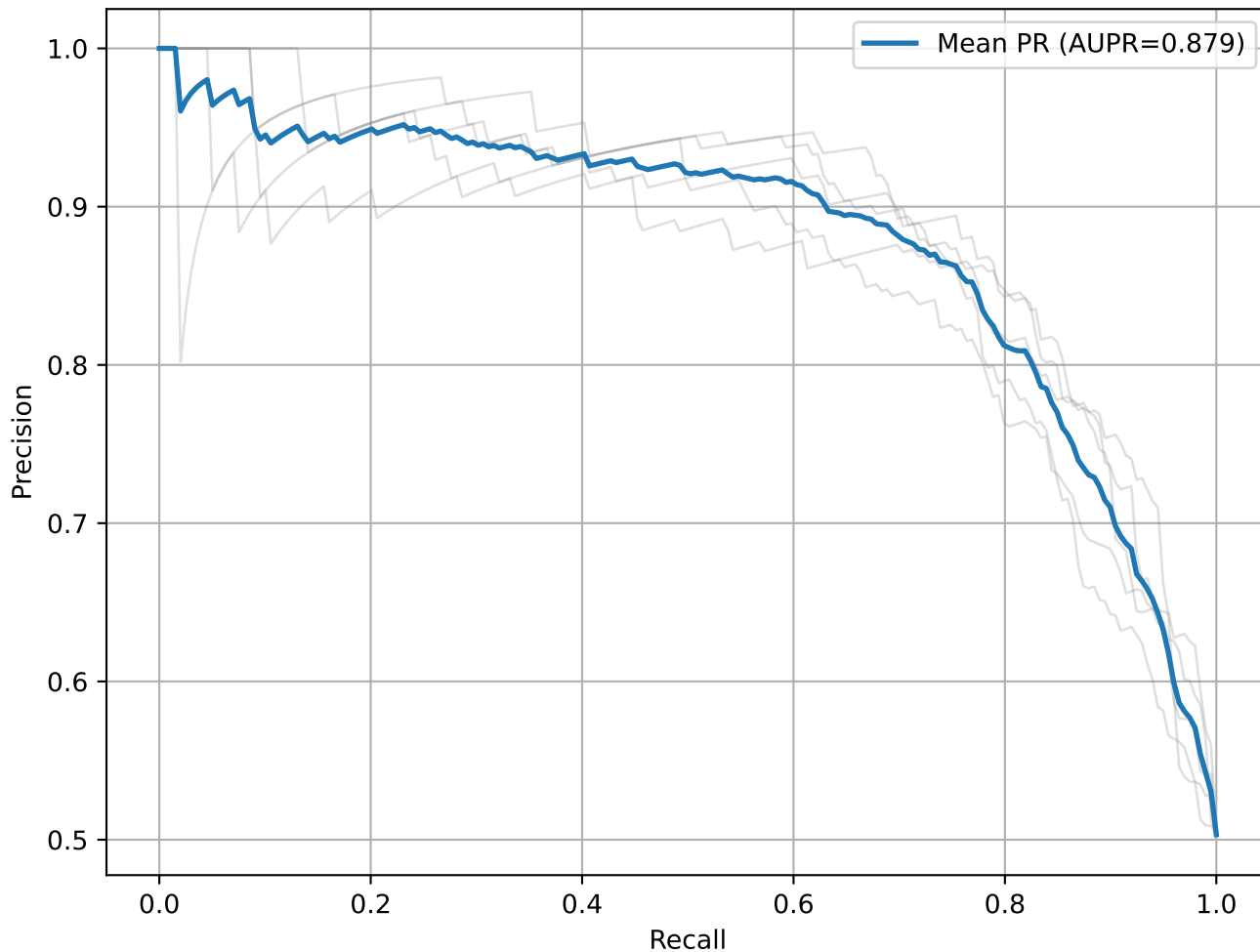
Classification Report - LGB (last fold)

	precision	recall	f1-score	support
0	0.87	0.85	0.86	199
1	0.86	0.88	0.87	202
accuracy			0.86	401
macro avg	0.86	0.86	0.86	401
weighted avg	0.86	0.86	0.86	401

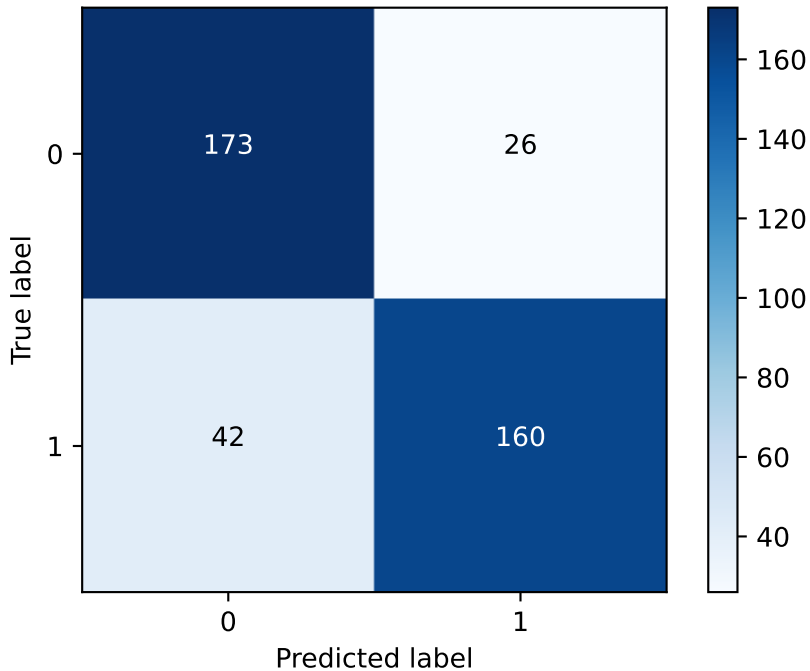
ROC - DNN



Precision-Recall - DNN



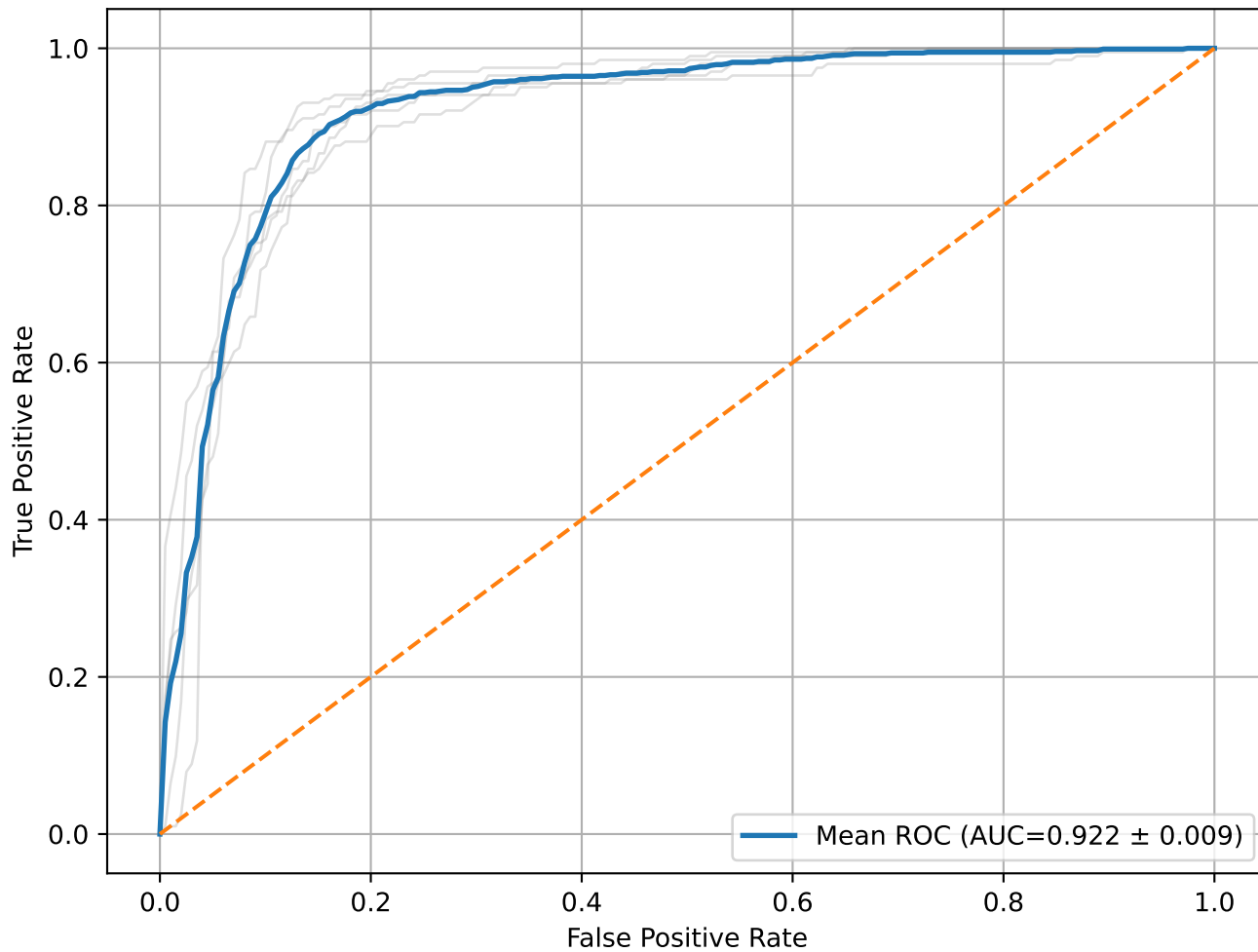
Confusion Matrix - DNN (last fold)



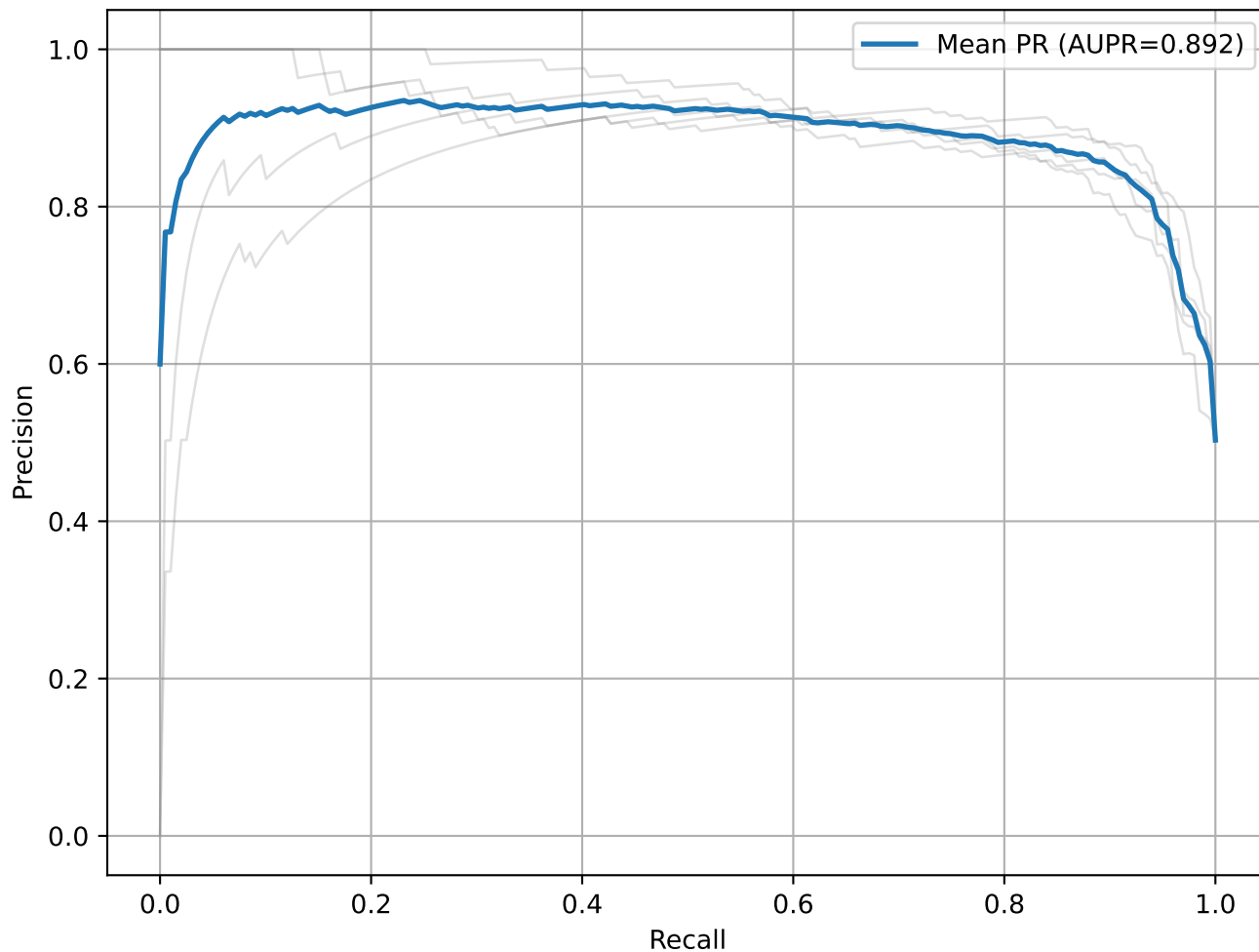
Classification Report - DNN (last fold)

	precision	recall	f1-score	support
0	0.80	0.87	0.84	199
1	0.86	0.79	0.82	202
accuracy			0.83	401
macro avg	0.83	0.83	0.83	401
weighted avg	0.83	0.83	0.83	401

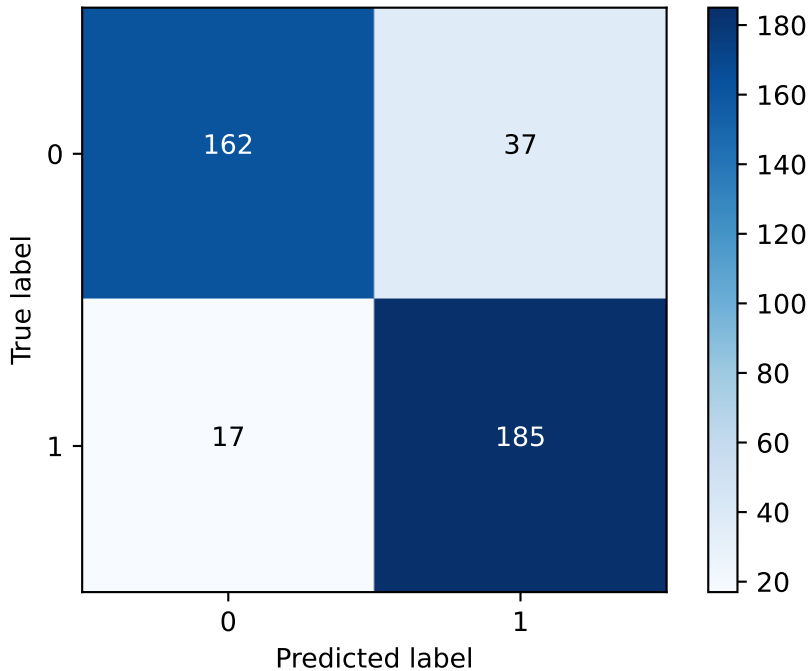
ROC - STACKING



Precision-Recall - STACKING



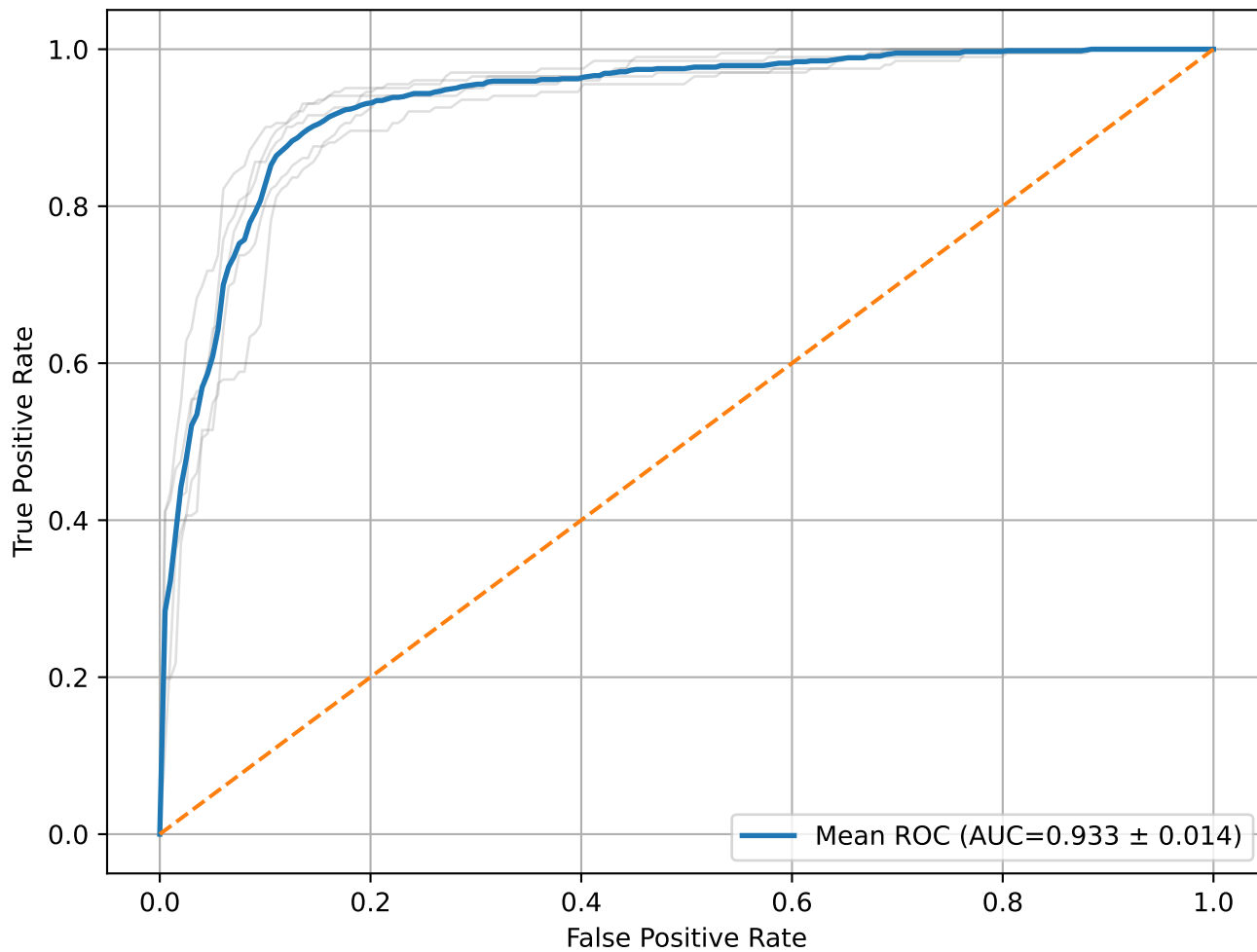
Confusion Matrix - STACKING (last fold)



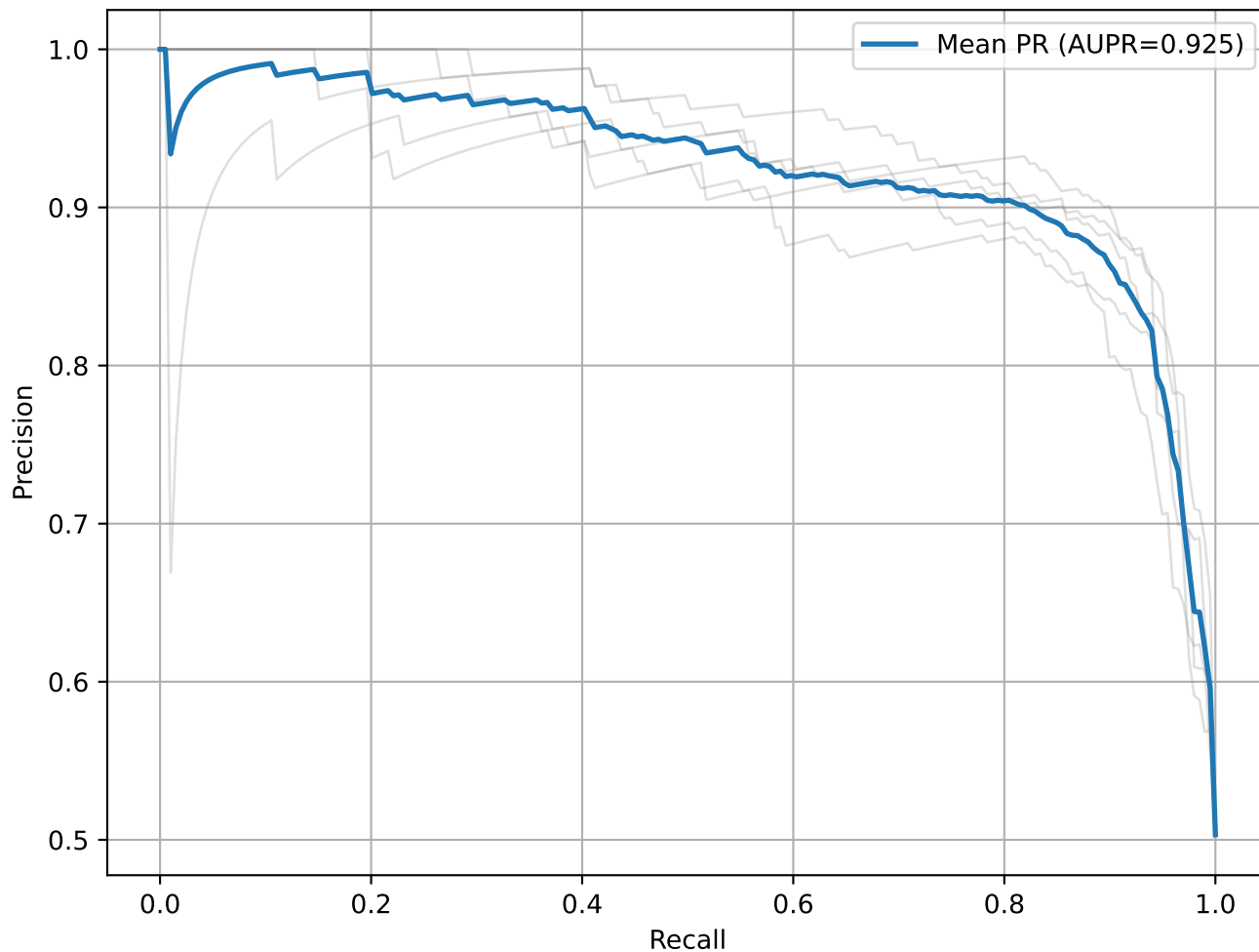
Classification Report - STACKING (last fold)

	precision	recall	f1-score	support
0	0.91	0.81	0.86	199
1	0.83	0.92	0.87	202
accuracy			0.87	401
macro avg	0.87	0.86	0.86	401
weighted avg	0.87	0.87	0.86	401

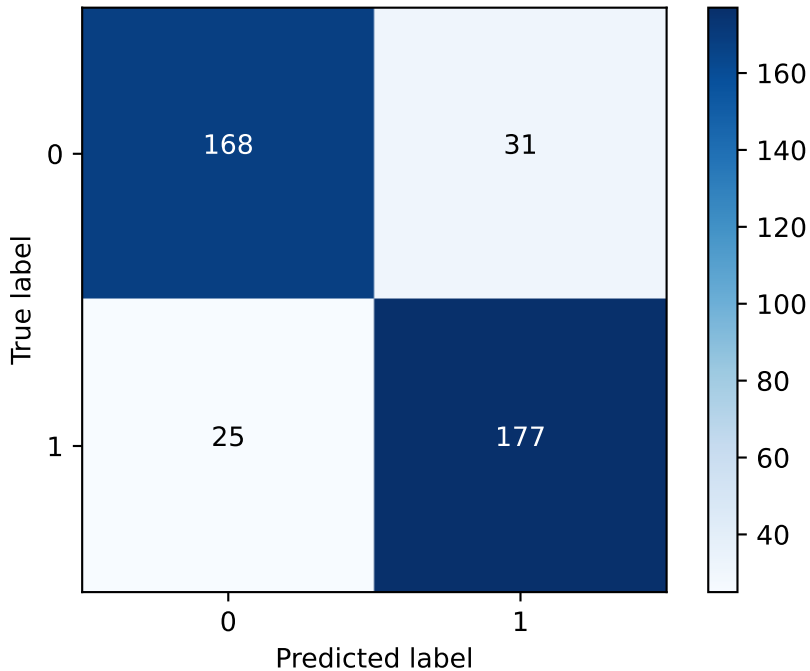
ROC - VOTING



Precision-Recall - VOTING



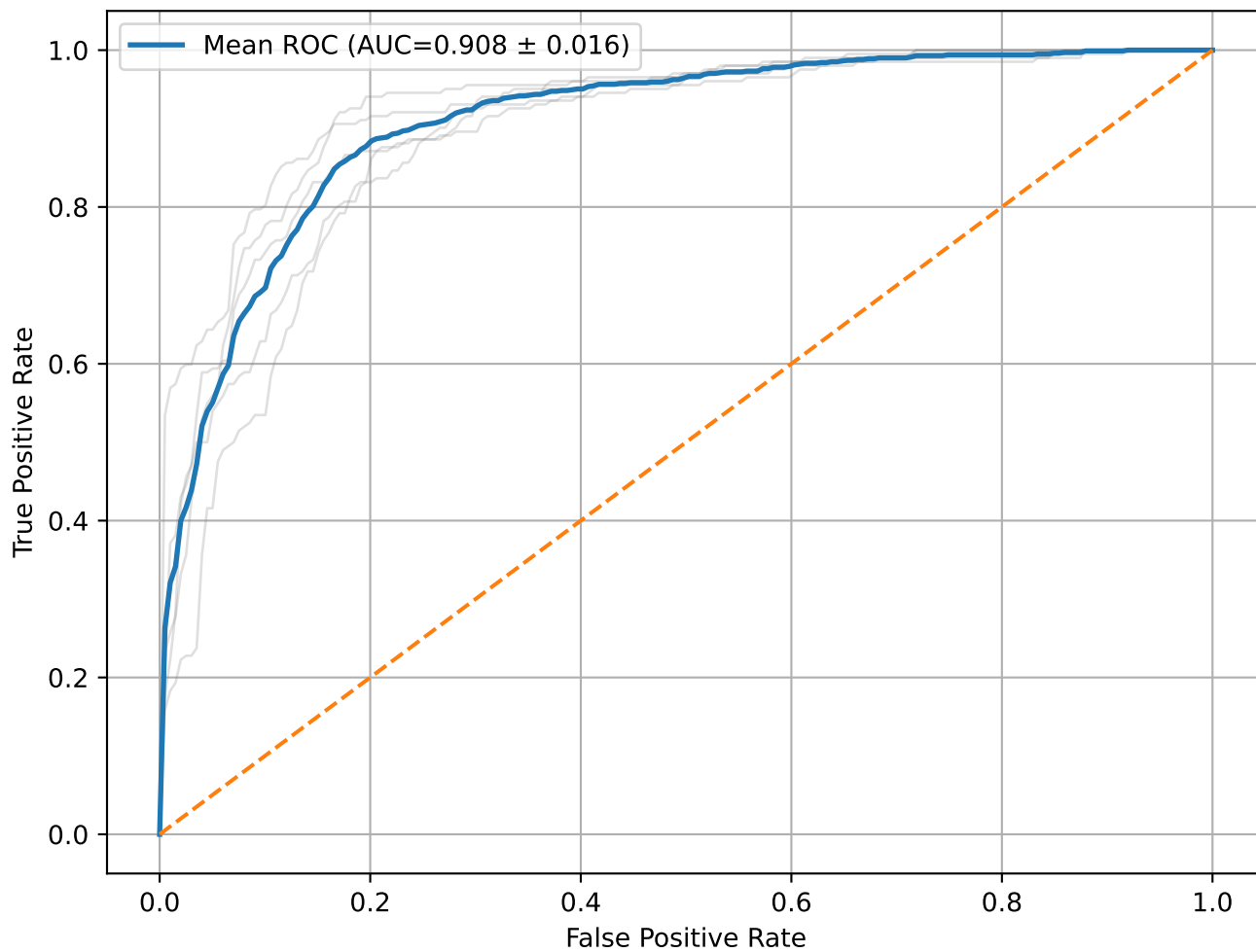
Confusion Matrix - VOTING (last fold)



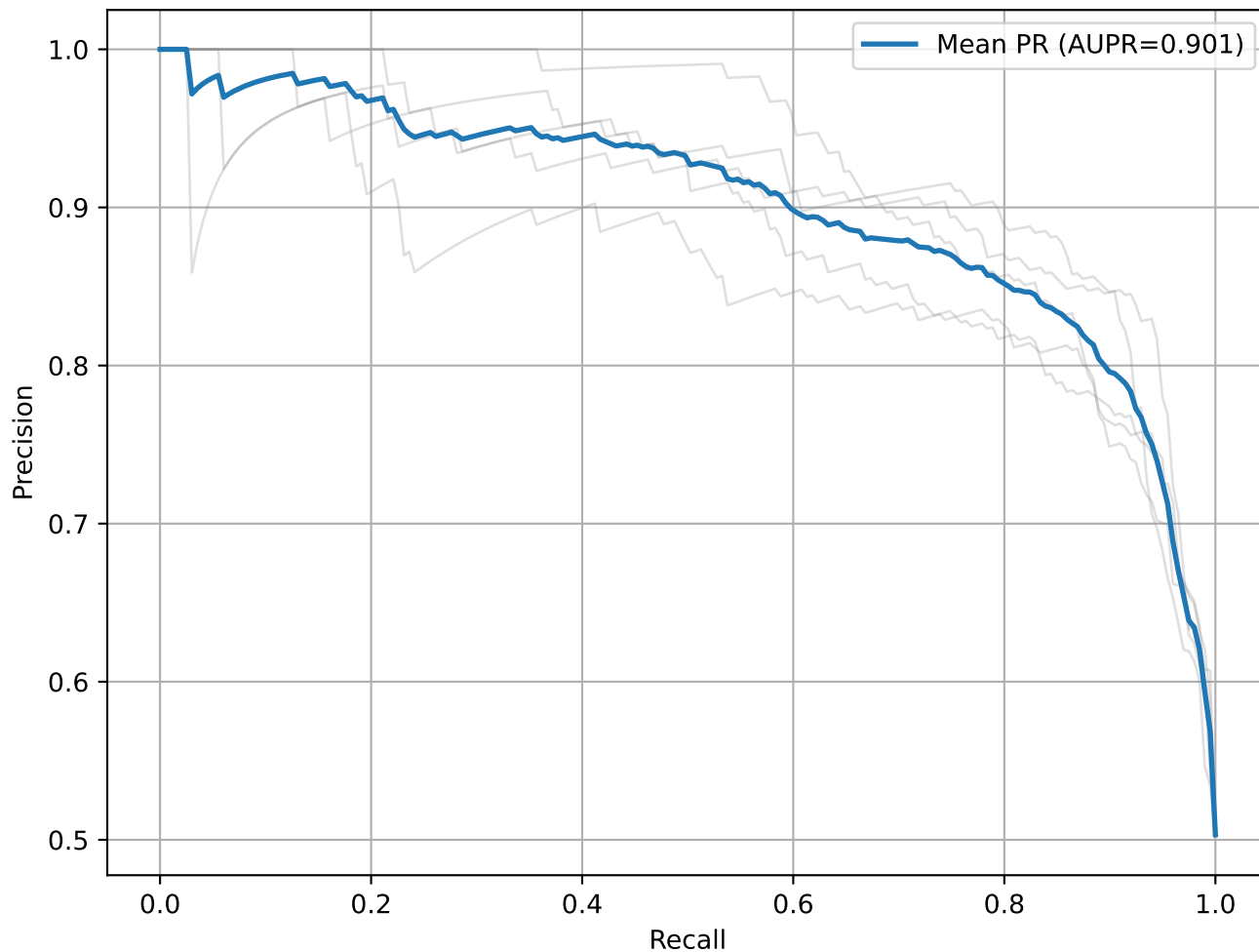
Classification Report - VOTING (last fold)

	precision	recall	f1-score	support
0	0.87	0.84	0.86	199
1	0.85	0.88	0.86	202
accuracy			0.86	401
macro avg	0.86	0.86	0.86	401
weighted avg	0.86	0.86	0.86	401

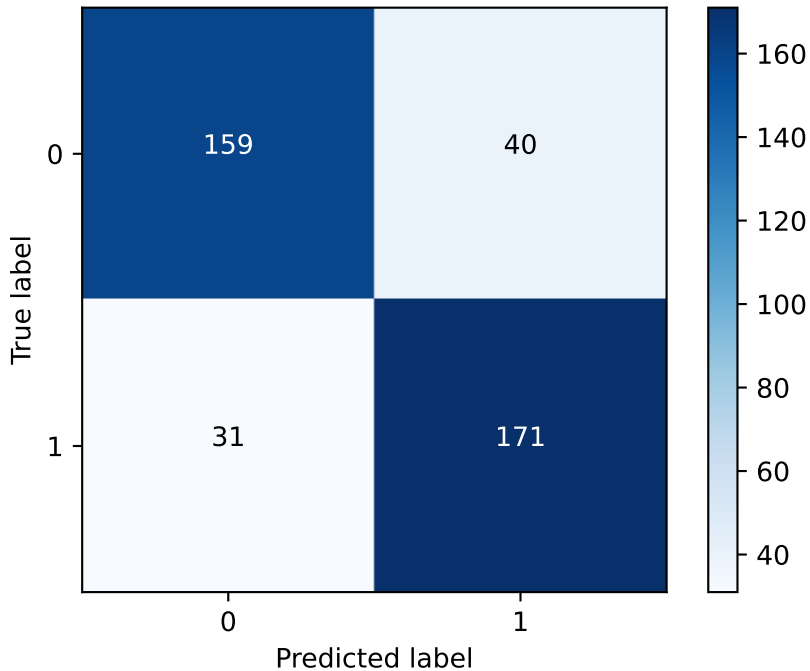
ROC - BAGGING



Precision-Recall - BAGGING



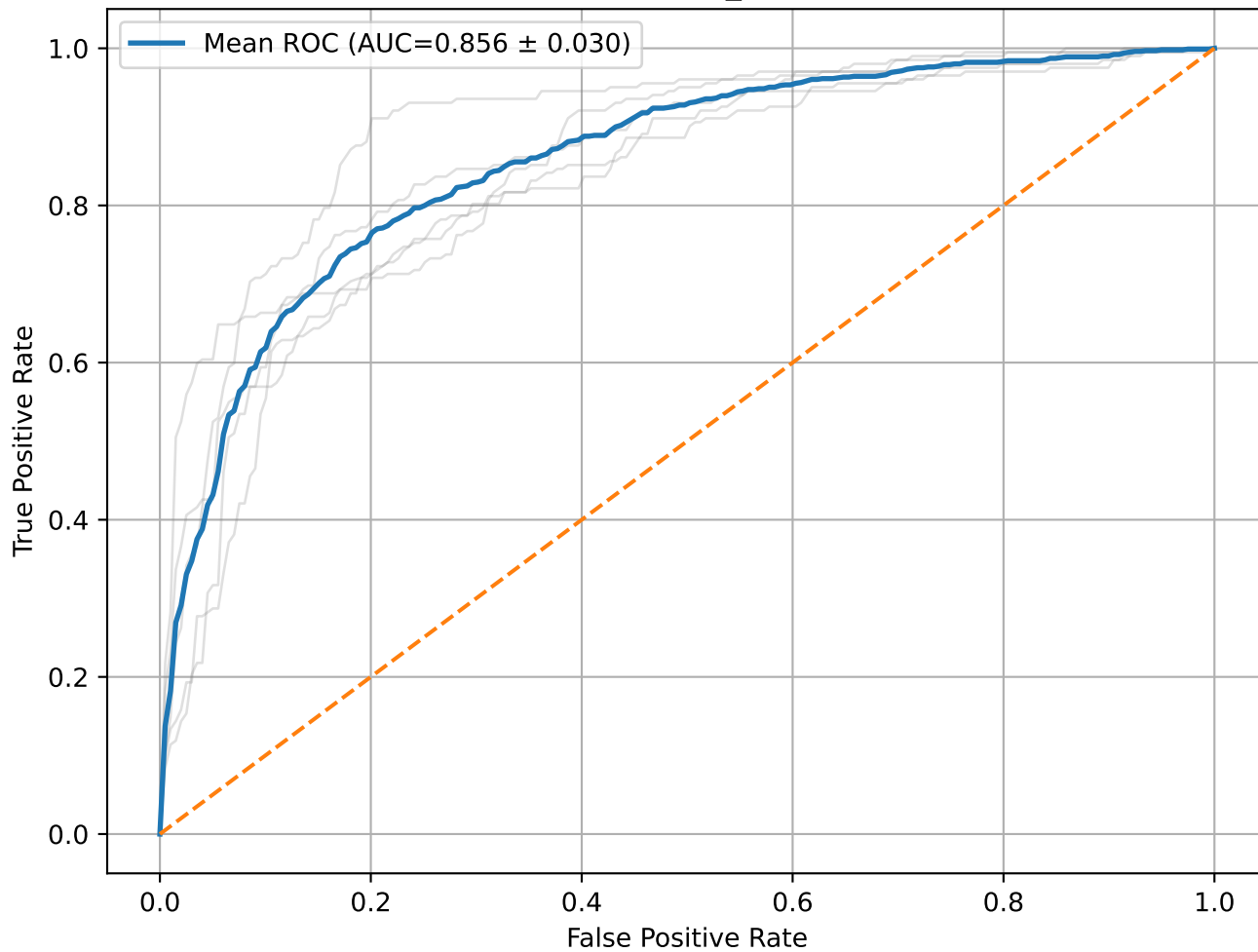
Confusion Matrix - BAGGING (last fold)



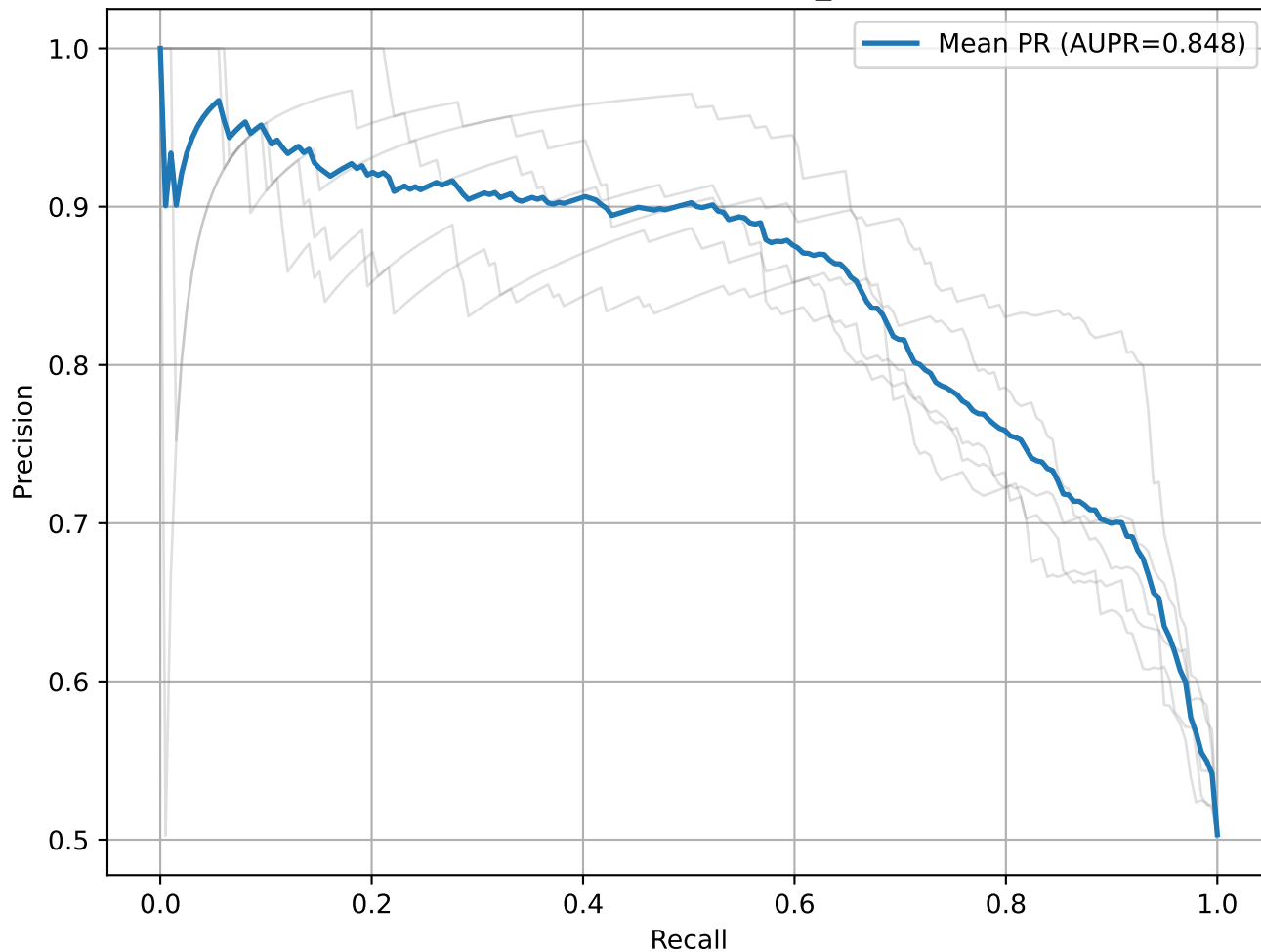
Classification Report - BAGGING (last fold)

	precision	recall	f1-score	support
0	0.84	0.80	0.82	199
1	0.81	0.85	0.83	202
accuracy			0.82	401
macro avg	0.82	0.82	0.82	401
weighted avg	0.82	0.82	0.82	401

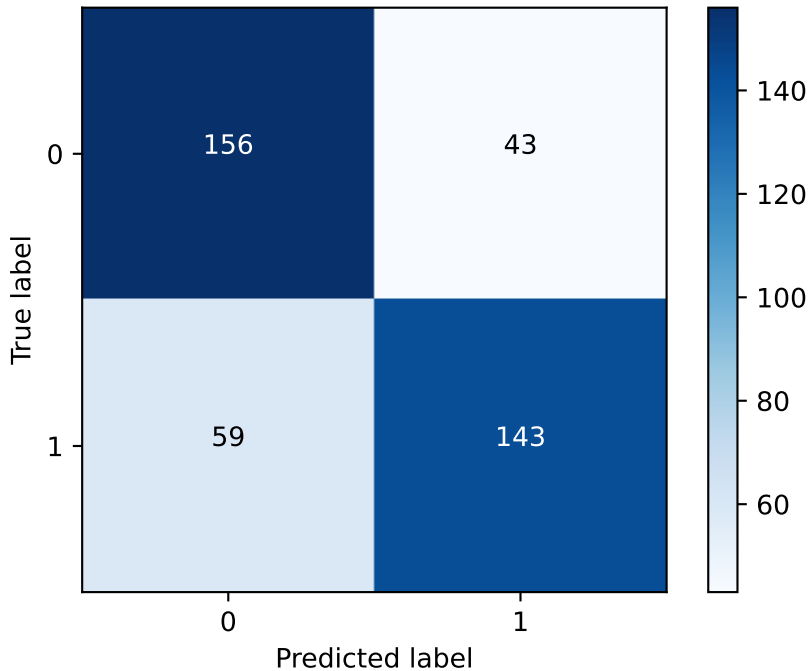
ROC - PCA_SVM



Precision-Recall - PCA_SVM



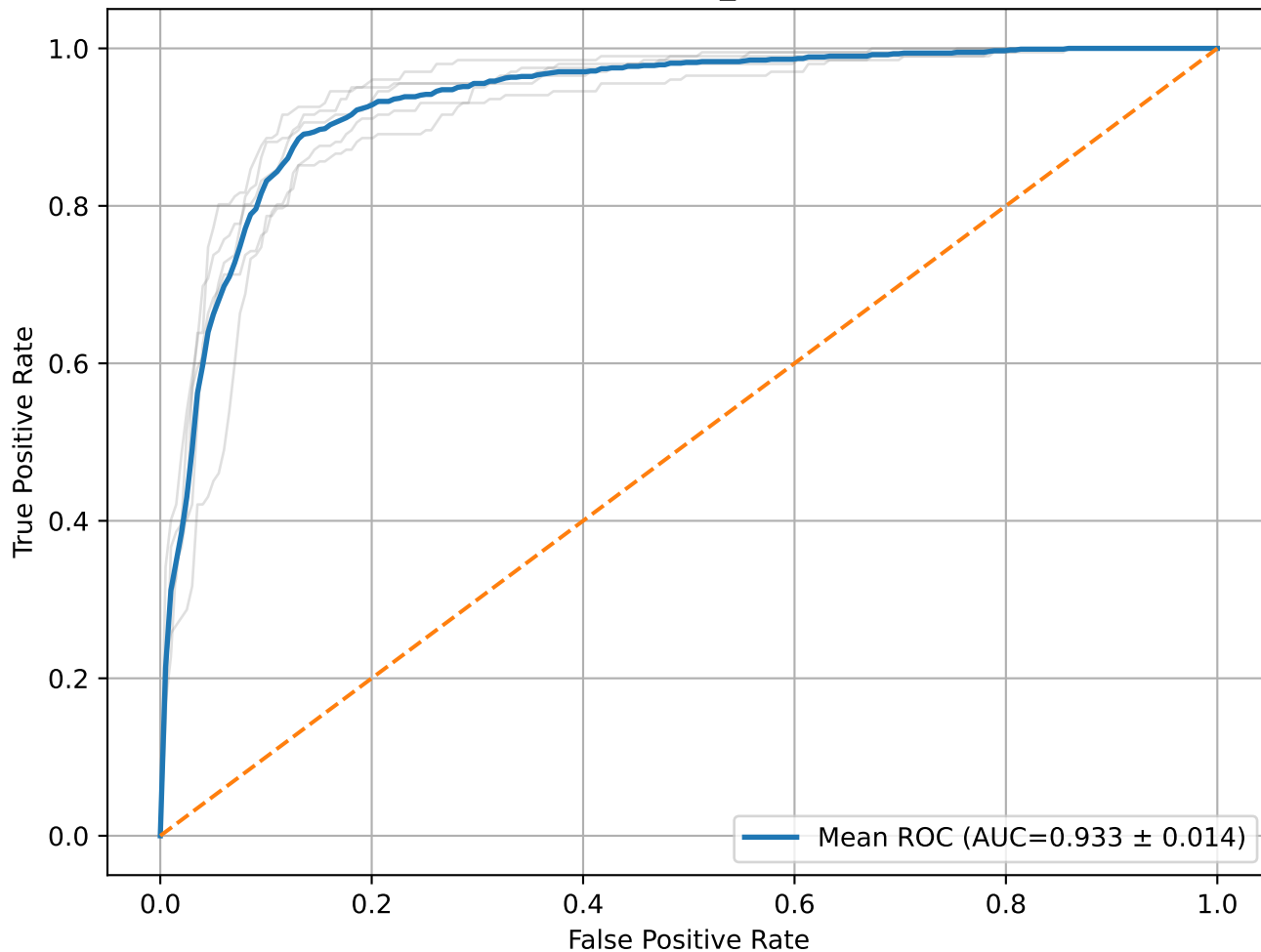
Confusion Matrix - PCA_SVM (last fold)



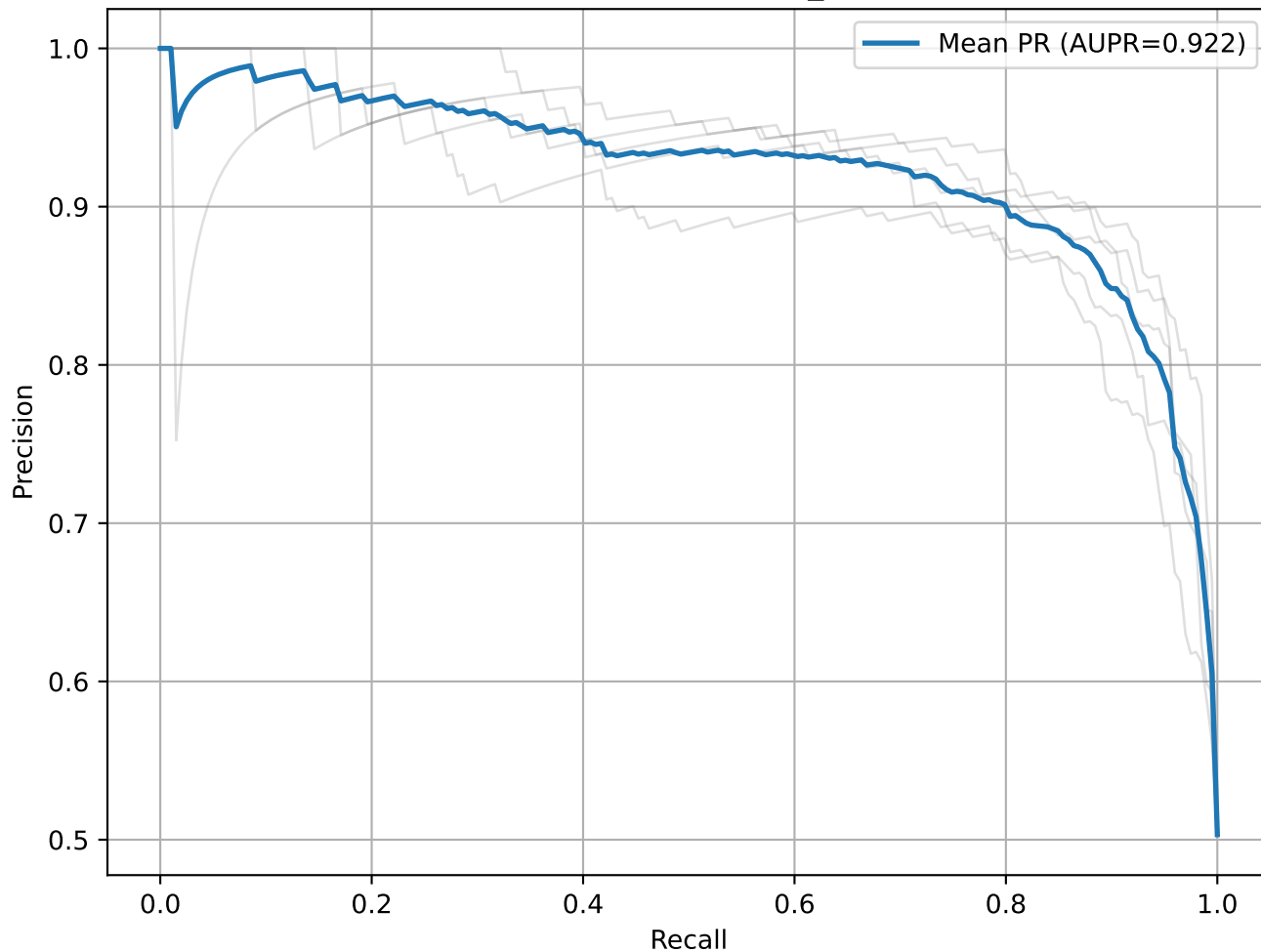
Classification Report - PCA_SVM (last fold)

	precision	recall	f1-score	support
0	0.73	0.78	0.75	199
1	0.77	0.71	0.74	202
accuracy			0.75	401
macro avg	0.75	0.75	0.75	401
weighted avg	0.75	0.75	0.75	401

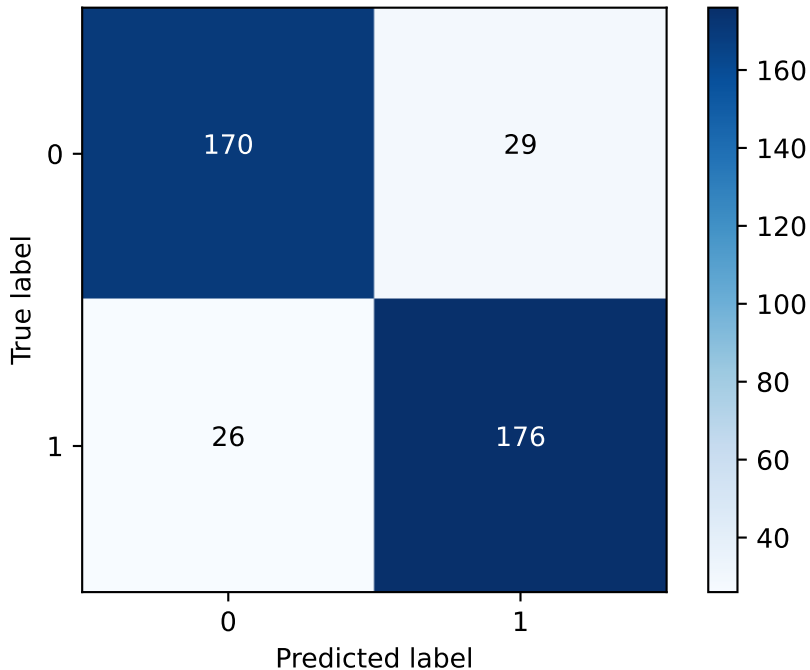
ROC - GA_XGB



Precision-Recall - GA_XGB



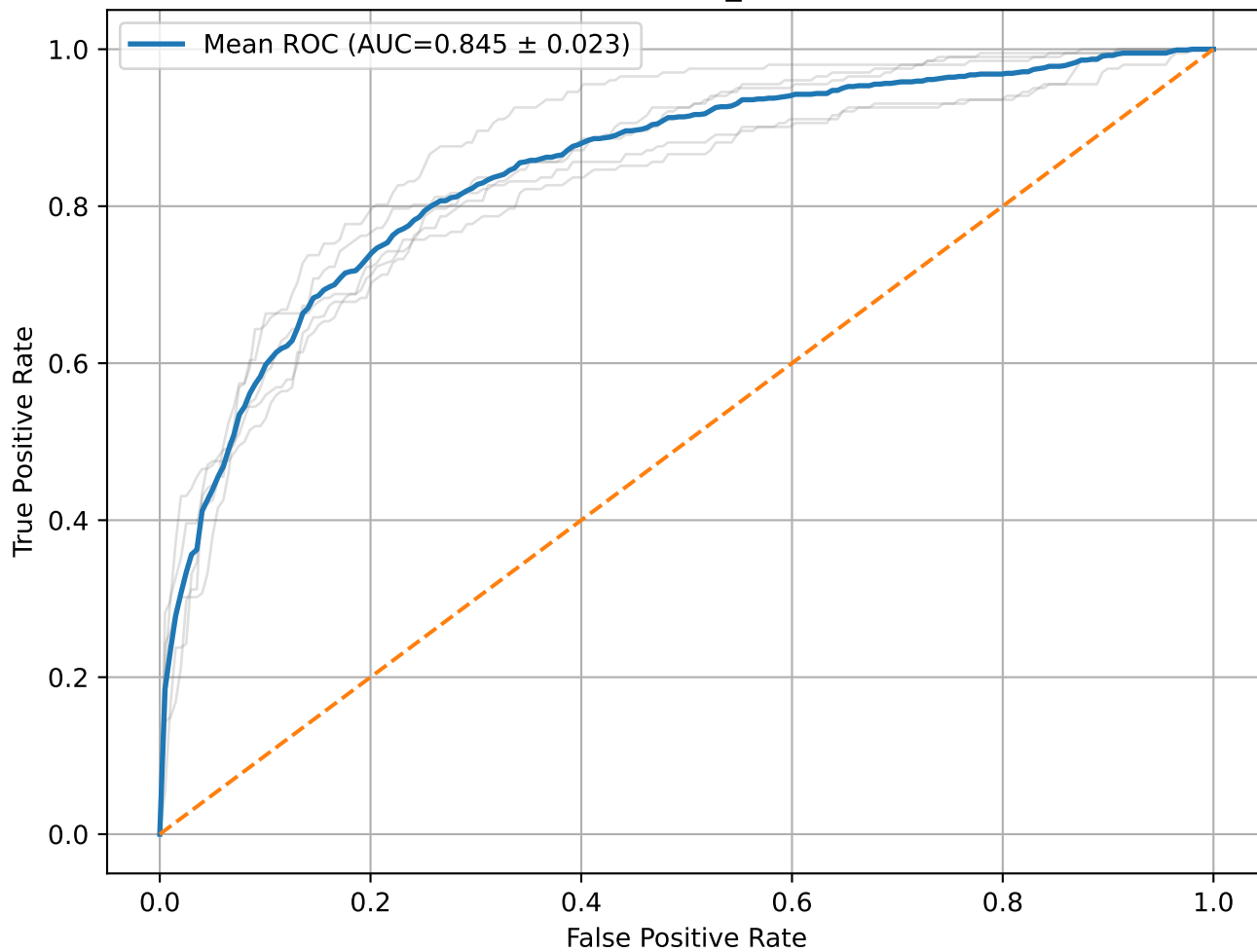
Confusion Matrix - GA_XGB (last fold)



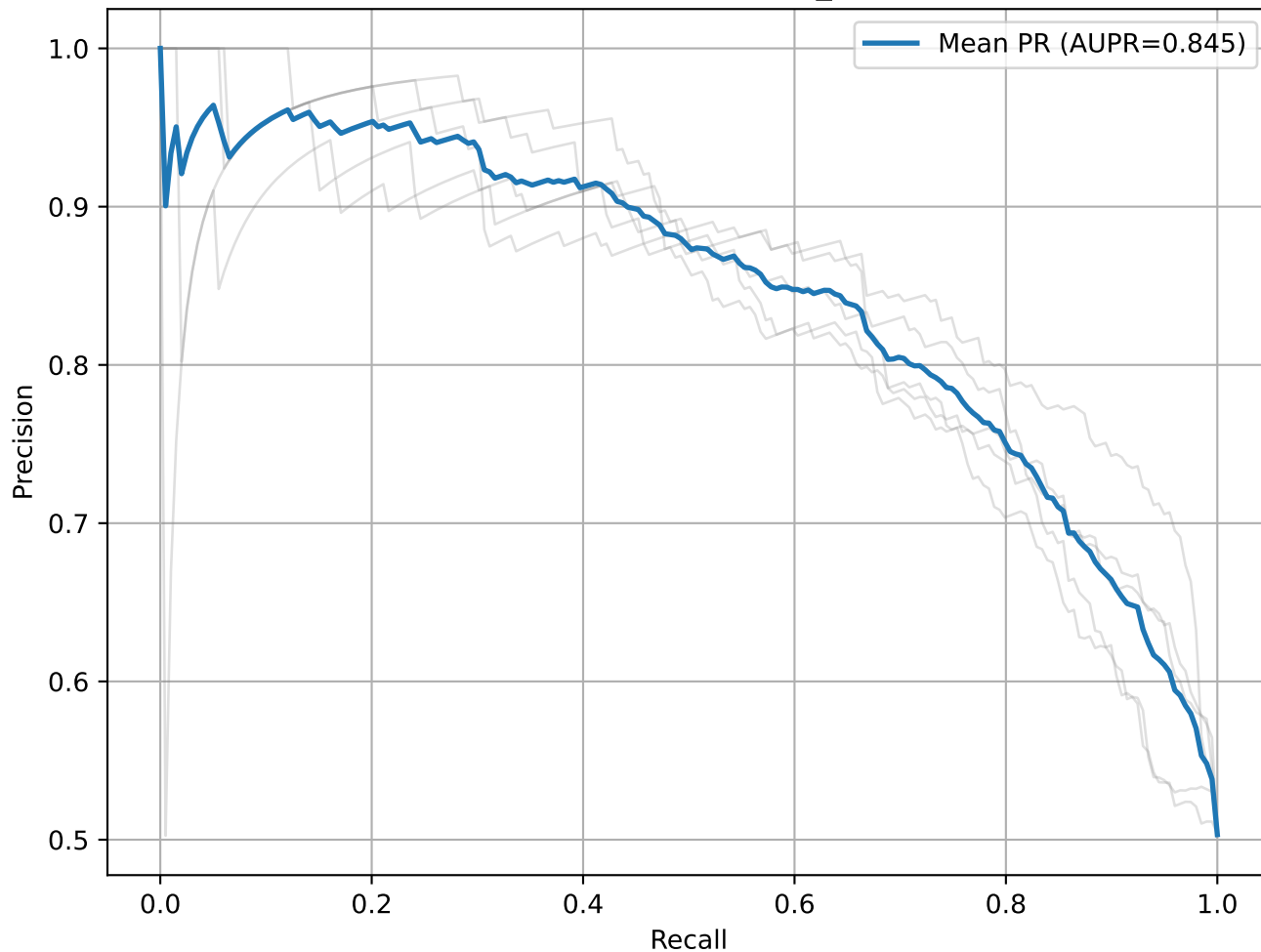
Classification Report - GA_XGB (last fold)

	precision	recall	f1-score	support
0	0.87	0.85	0.86	199
1	0.86	0.87	0.86	202
accuracy			0.86	401
macro avg	0.86	0.86	0.86	401
weighted avg	0.86	0.86	0.86	401

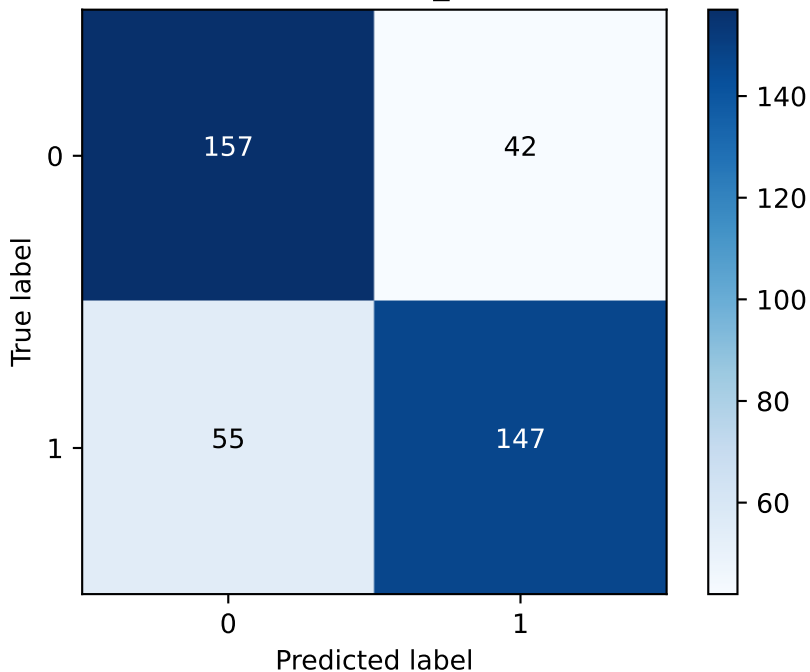
ROC - PSO_SVM



Precision-Recall - PSO_SVM



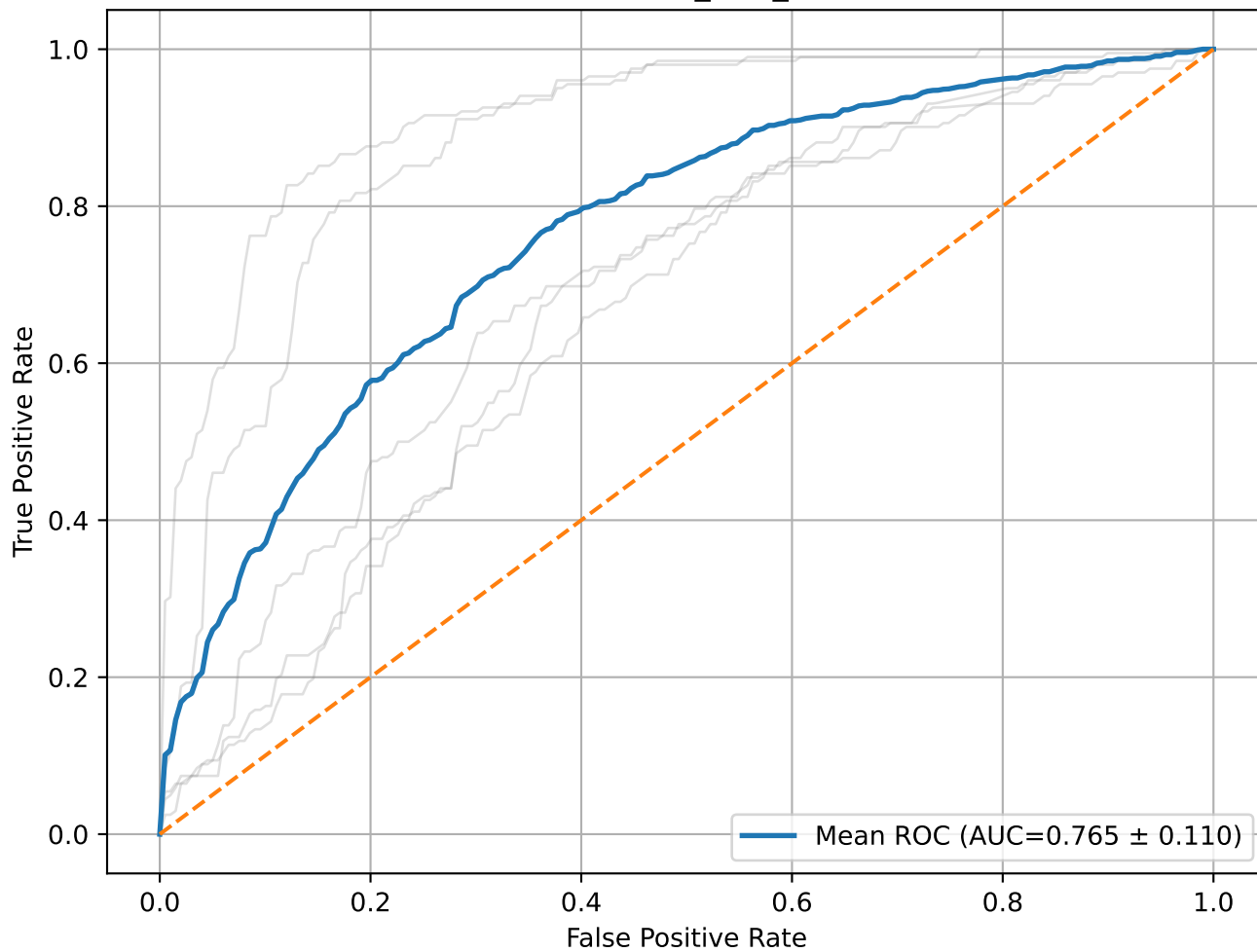
Confusion Matrix - PSO_SVM (last fold)



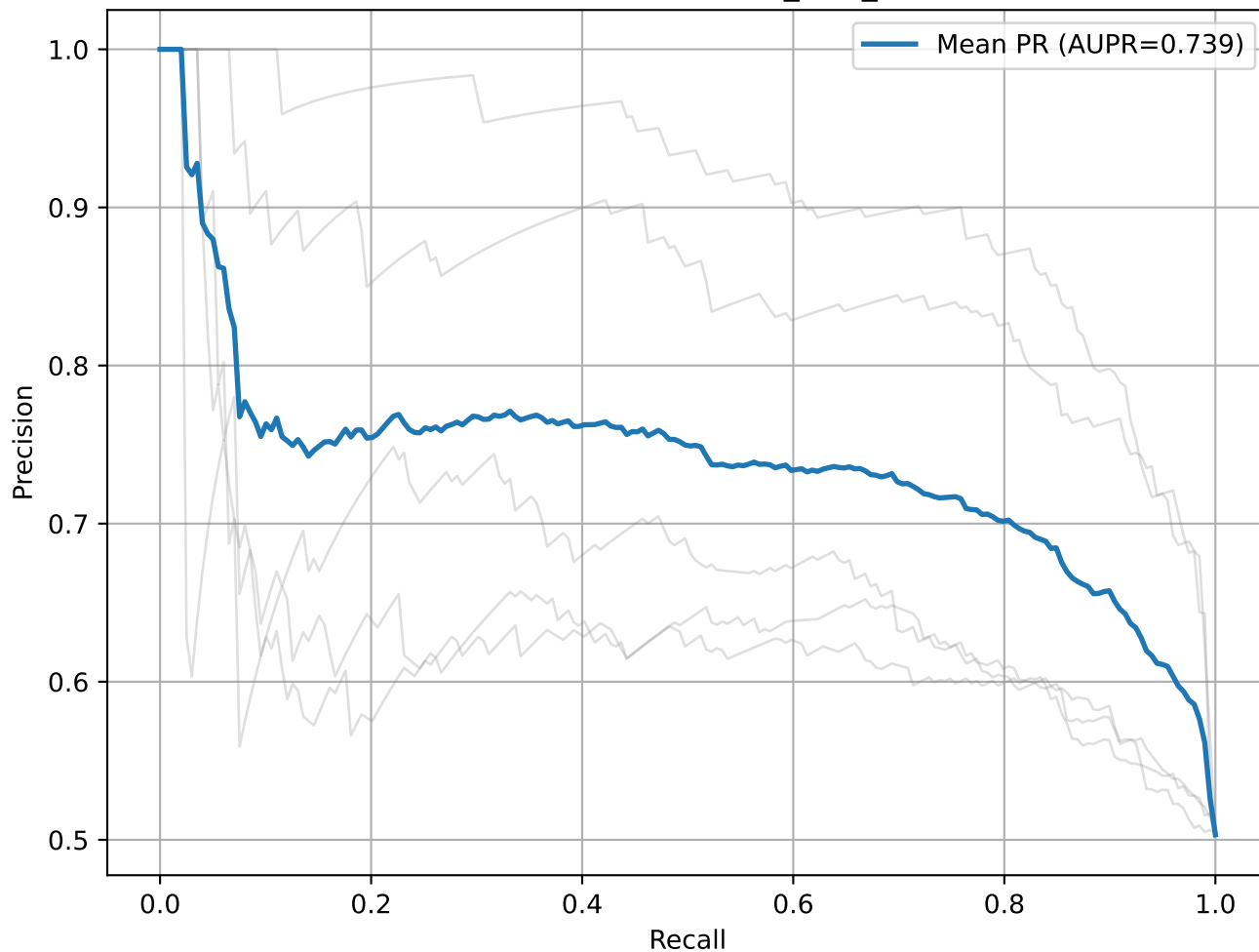
Classification Report - PSO_SVM (last fold)

	precision	recall	f1-score	support
0	0.74	0.79	0.76	199
1	0.78	0.73	0.75	202
accuracy			0.76	401
macro avg	0.76	0.76	0.76	401
weighted avg	0.76	0.76	0.76	401

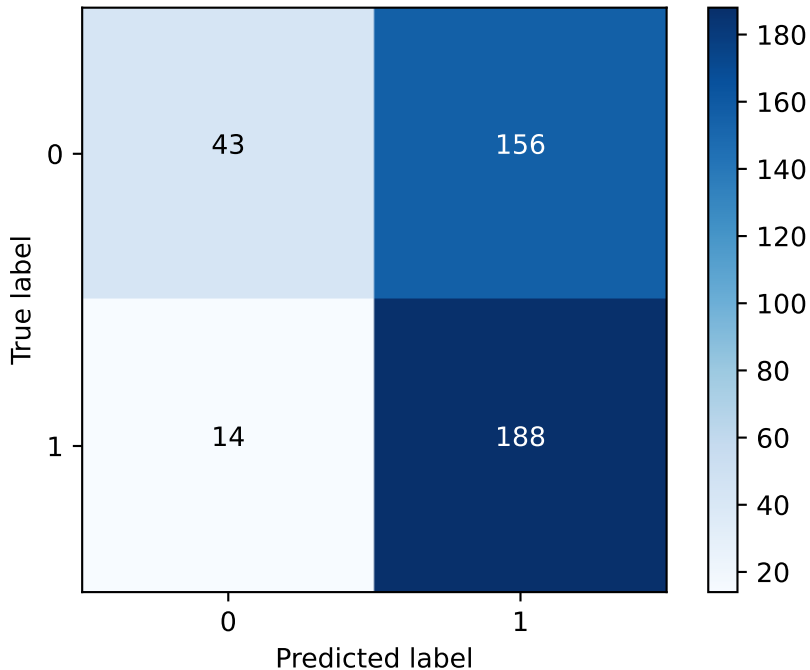
ROC - BAYES_OPT_SVMS



Precision-Recall - BAYES_OPT_SVMS



Confusion Matrix - BAYES_OPT_SVMS (last fold)



Classification Report - BAYES_OPT_SVMS (last fold)

	precision	recall	f1-score	support
0	0.75	0.22	0.34	199
1	0.55	0.93	0.69	202
accuracy			0.58	401
macro avg	0.65	0.57	0.51	401
weighted avg	0.65	0.58	0.51	401

Model Comparison Table (sorted by mean ROC AUC)

model	mean_roc_auc	std_roc_auc	mean_aupr	mean_accuracy	mean_f1
lgb	0.938077	0.011870	0.930422	0.876925	0.879549
ga_xgb	0.931494	0.014026	0.922083	0.874931	0.876832
xgb	0.931262	0.011221	0.920409	0.868948	0.871132
voting	0.931251	0.013817	0.924500	0.878915	0.881333
rf	0.929757	0.016295	0.920556	0.878411	0.880483
stacking	0.919770	0.008698	0.891477	0.870453	0.876149
bagging	0.906195	0.016112	0.900189	0.839054	0.840440
logistic	0.896748	0.012756	0.879431	0.827603	0.830463
dnn	0.877206	0.014863	0.878460	0.815142	0.808349
pca_svm	0.853896	0.030215	0.847635	0.771798	0.760341
pso_svm	0.842993	0.022818	0.844794	0.762347	0.743348
svm	0.833083	0.024587	0.842738	0.753365	0.730912
bayes_opt_svms	0.763069	0.110449	0.739224	0.678608	0.746555

Combined ROC - All Models

